

Spatial networks and inequalities

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Economic Complexity, Class 7 – Nov 11 2025

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Quantifying the Influence of Climate on Human Conflict

Research Question: *To what degree do spatially concentrated changes in climate affect the intensity and distribution of conflict between people across different regions, time periods, and types of violence? In other words, are deviations in climate consistently associated with increased risk of human conflict?*

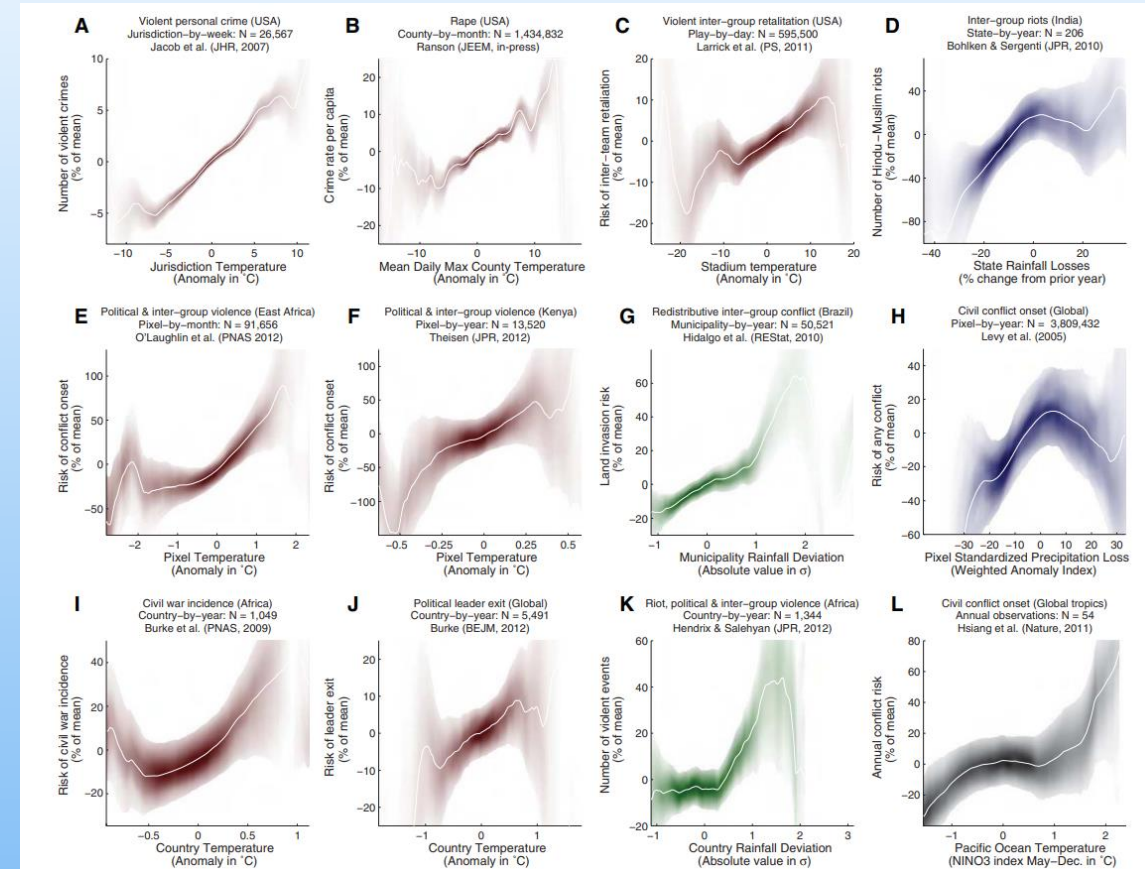
The Data:

- 60 primary studies, 45 different conflict data sets ultimately examined
- Time periods: 10,000 BCE to 2013 (12,000 years)
- Types of human conflict: interpersonal/local (city-level crime), intergroup/subnational (country or province), institutional/national, civilization/global

Method: The research design was a meta-analysis of quantitative studies linking changes in climate (mainly temperature and precipitation anomalies) to intensity of human conflict. For a study to be included in the analysis, it must quantitatively link climate variables to conflict outcomes, have a causal or quasi-causal relationship (not just correlational), and must have standard errors or sufficient statistics to calculate effect sizes. They

standardized all climate coefficients that showed how much conflict risk increases per one standard deviation increase in climate shift. Then, they used weighted least squares to estimate average effect size, differing effects across conflict types, and variations by time scale. Effects across spatial scales were also considered here.

Findings: A one standard deviation increase in temperature or precipitation anomaly leads to (on average) a 4% increase in local violence and a 14% increase in subnational conflict. The effects were greatest in more densely populated areas like Africa, South Asia, and Latin America. The relationship is statistically significant and potentially large under anticipated future global warming. *This suggests that spatial concentration of people in climate-sensitive regions could increase the risk of urban conflict.*



The second block

3	Oct 7	Balázs Lengyel Knowledge, innovation, and technological development	II. Innovation and Economic Geography	[4]
4	Oct 14	Balázs Lengyel The development of cities and regions: agglomeration economies, the specialization-diversity debate		[5,6]
5	Oct 21	Zoltán Elekes Labor markets and skills: labor mobility, skill-relatedness		[7,8]
6	Nov 4	Balázs Lengyel Learning and innovation in networks: knowledge sourcing and dynamics of networks		[9,10]
7	Nov 11	Balázs Lengyel and Sándor Juhász Spatial and network inequalities: social capital, segregation and mixing		[11,12]
8	Nov 18	Presentation of students' research proposal		

[4] Dosi, G. (1988). Sources, procedures, and microeconomic effects of innovation. *Journal of Economic Literature*, 1120-1171.

[5] Audretsch, D. B., & Feldman, M. P. (1996). R&D spillovers and the geography of innovation and production. *The American Economic Review*, 86(3), 630-640.

[6] Glaeser, E. L., Kallal, H. D., Scheinkman, J. A., & Shleifer, A. (1992). Growth in cities. *Journal of Political Economy*, 100(6), 1126-1152.

[7] Neffke, F., Henning, M., & Boschma, R. (2011). How do regions diversify over time? Industry relatedness and the development of new growth paths in regions. *Economic Geography*, 87(3), 237-265.

[8] Alabdulkareem, A., Frank, M. R., Sun, L., AlShebli, B., Hidalgo, C., & Rahwan, I. (2018). Unpacking the polarization of workplace skills. *Science Advances*, 4(7), eaao6030.

[9] Sorenson, O., Rivkin, J. W., & Fleming, L. (2006). Complexity, networks and knowledge flow. *Research Policy*, 35(7), 994-1017.

[10] Abbasiharofteh, M., Kogler, D. F., & Lengyel, B. (2023). Atypical combinations of technologies in regional co-inventor networks. *Research Policy*, 52(10), 104886.

[11] Tóth, G., Wachs, J., Di Clemente, R., Jakobi, Á., Ságvári, B., Kertész, J., & Lengyel, B. (2021). Inequality is rising where social network segregation interacts with urban topology. *Nature Communications*, 12(1), 1143.

[12] Aiello, L. M., Vybornova, A., Juhász, S., Szell, M., & Bokányi, E. (2025). Urban highways are barriers to social ties. *PNAS*, 122(10), e2408937122.



Why do inequalities grow in a small world?

Network fragmentation and inequalities

Small-world social networks are fragmented by groups.

One of the strongest forces of group formation is prosperity.

Diffusion of innovation is unequal across network fragments.



Today

1. Social network , residential and experienced segregation
2. Network fragmentation, urban topology and inequalities
3. Social media and urban inequalities / Sándor
4. Homeworks and assignments

1. Social capital, residential and experienced segregation

Sarkar, S., Cottineau-Mugadza, C., & Wolf, L. J. (2024). Spatial inequalities and cities: A review. *Environment and Planning B: Urban Analytics and City Science*, 51(7), 1391-1407.

Research Question: The aim is to focus on the spatial-geographic dimension of how inequalities play out in urban space, and to identify and organise what it is about cities that fuels inequalities.

Data: 17 empirical papers, across 9 countries and several hundred cities.

- OpenStreetMap (OSM) – street networks & transit
- Mobile phone data – mobility & accessibility
- Census/microdata – demographics & income

Methods: Summary of methodologies used in the analyzed articles, compare them, and identify gaps.

Built their own framework to help distinguish between simple spatial differences and deeper forms of structural unfairness, pushing future research toward normative, justice-oriented analysis

Conceptual gradient: Disparity → Inequality → Injustice

Key Findings: Identify key factors that shape inequalities: (1) **Location**, (2) **Movement**, (3) **Infrastructure**

High-income groups enjoy better accessibility and shorter travel times.

Urban sprawl increases energy, transport, and health inequalities.

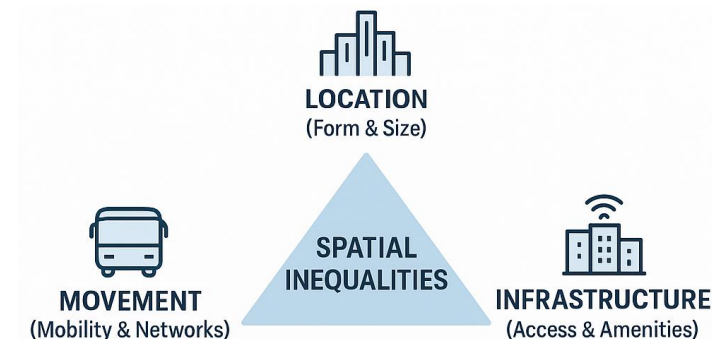
Transit-oriented and compact urban forms improve accessibility and sustainability.

Urban structure and transport networks can be powerful levers to reduce inequality.

Future urban science should: Integrate AI/ML responsibly,

Strengthen normative (justice-based) approaches,

Emphasize public participation and equitable policy design.



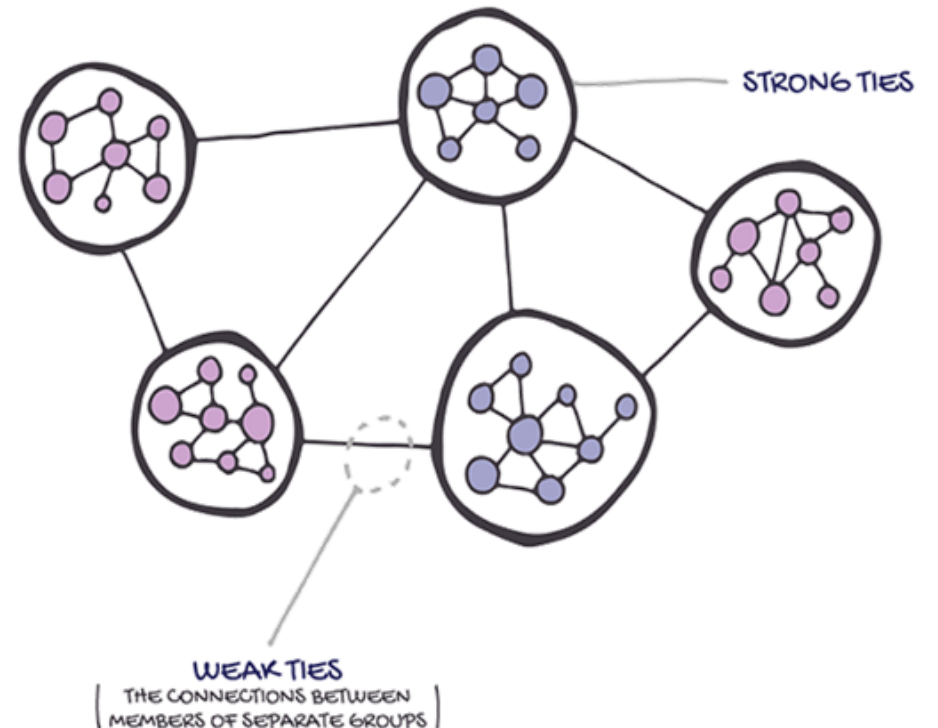
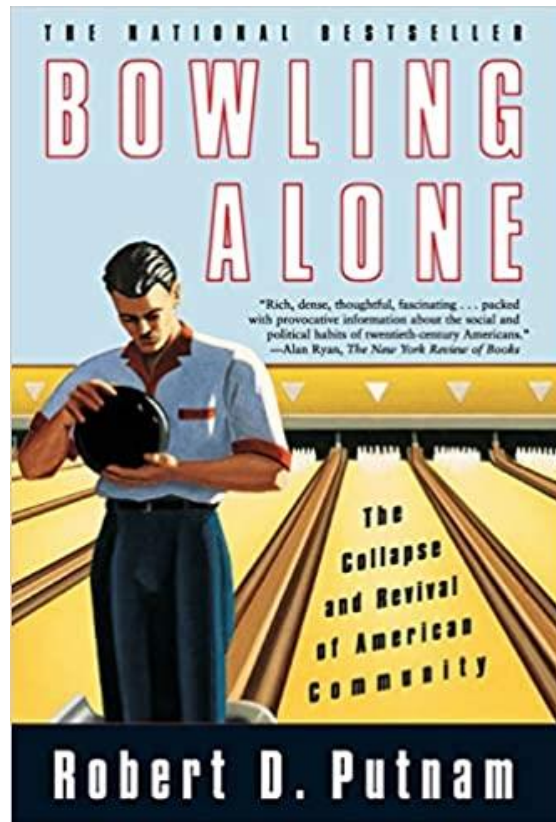
Disparity → Inequality → Injustice

Social capital

The to mobilize resources through social contacts. (Putnam)

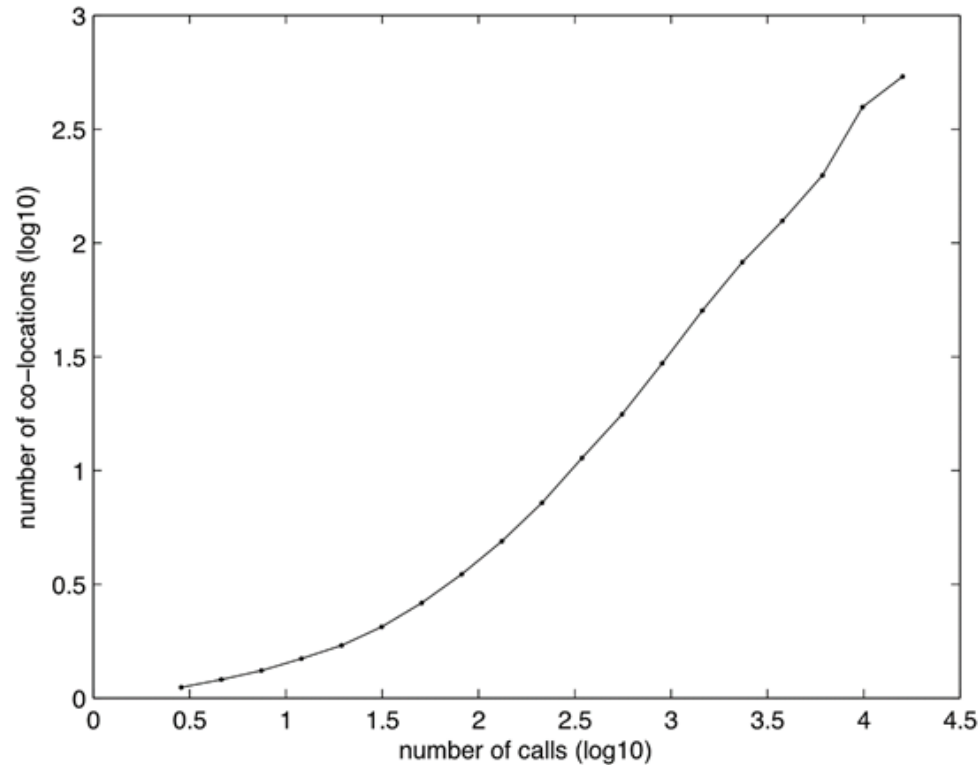
Bonding social capital: strong connections (Coleman)

Bridging social capital: weak tie (Granovetter)

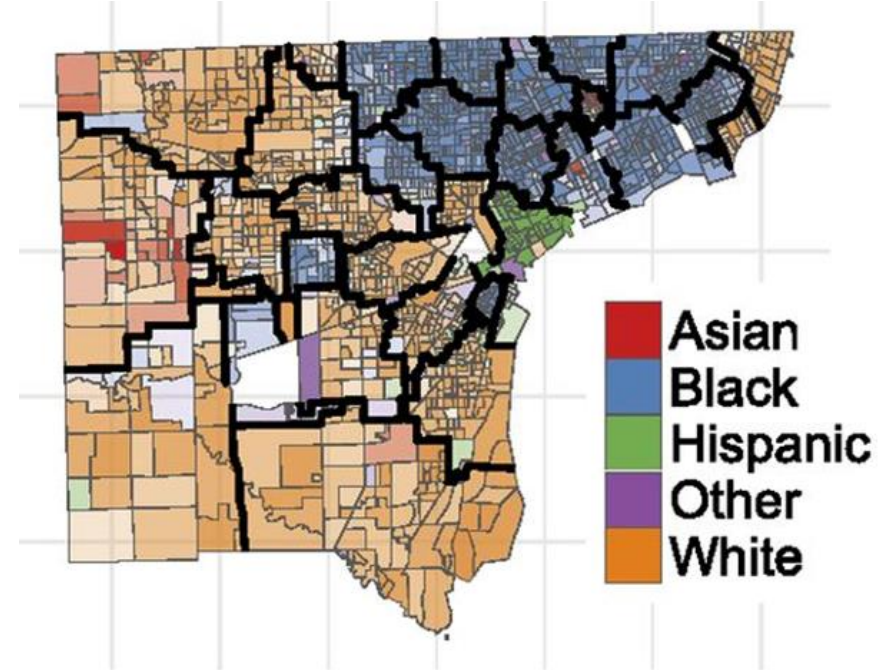


Social capital and social networks in cities

Geographic constraints of social ties increase segregation



Calabrese-Smoreda-Blondel-Ratti (2011) PLOS ONE

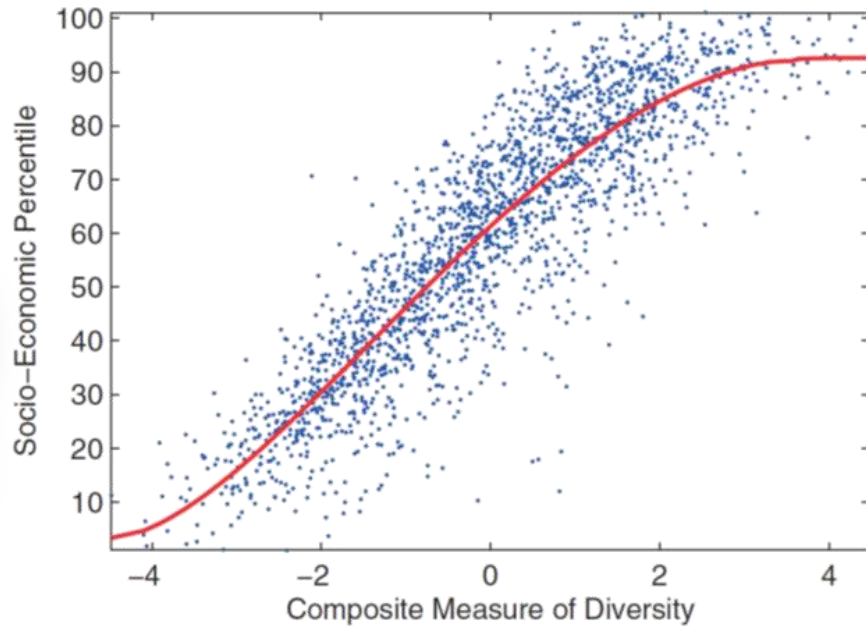


(a) Detroit, $k = 34$

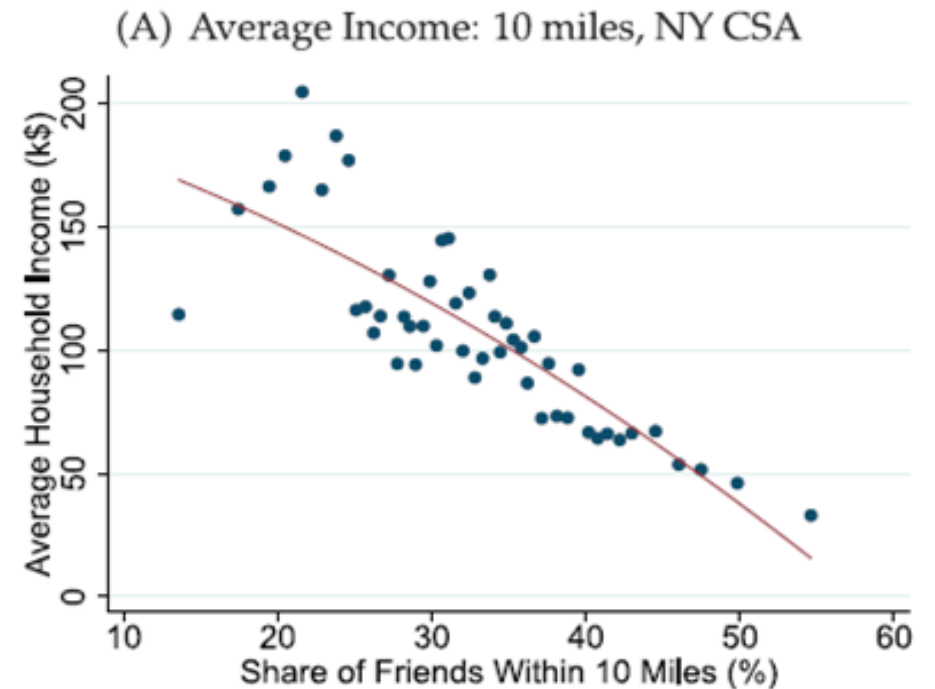
Philip S. Chodrow PNAS 2017;114:44:11591-11596

Social capital and social networks in cities

Geographic constraints of social ties predict socio-economic status



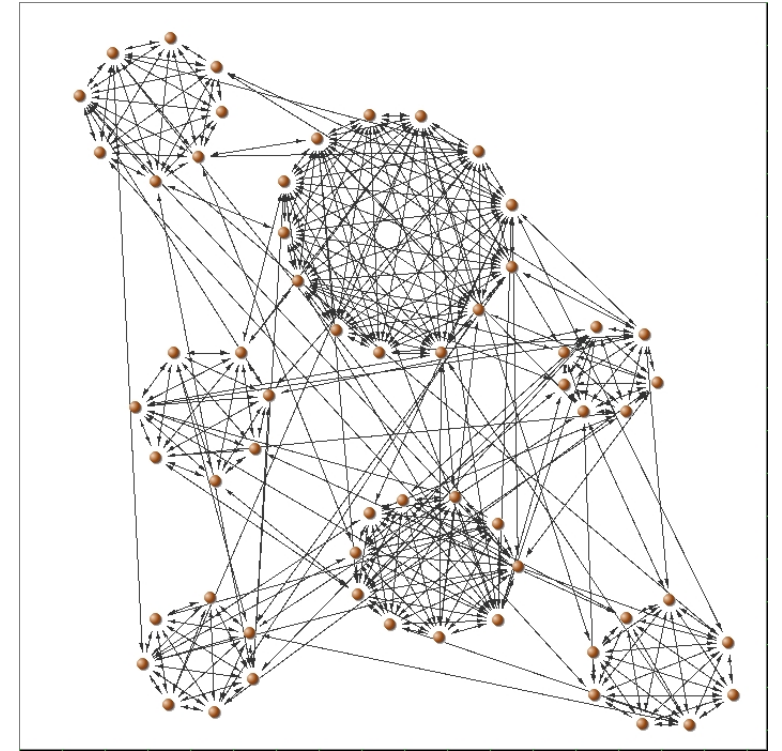
Eagle-Macy-Claxton (2010) Science



Bailey et al. (2020) JUE

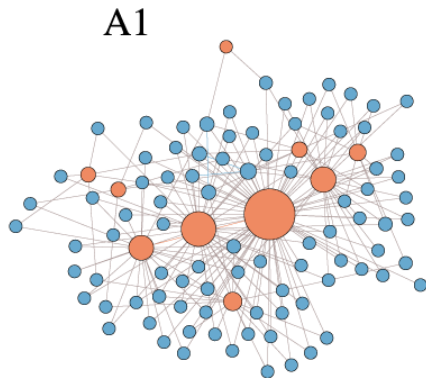
Homophily in small worlds

- Social networks cannot be described by random ties:
 - Dense local relations and
 - few bridges between isolated groups
- Small worlds
 - High clustering – due to dense local networks
 - Short paths – due to bridges

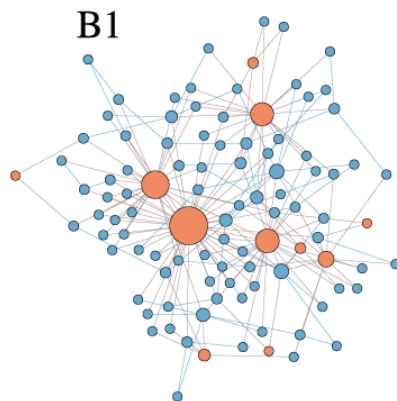


complete heterophily

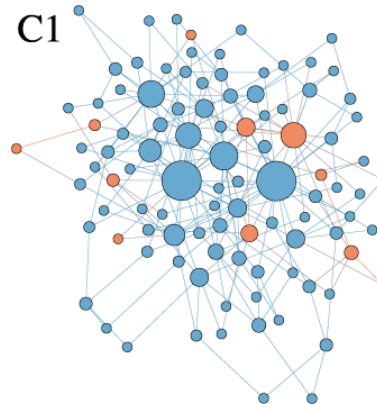
$h = 0$



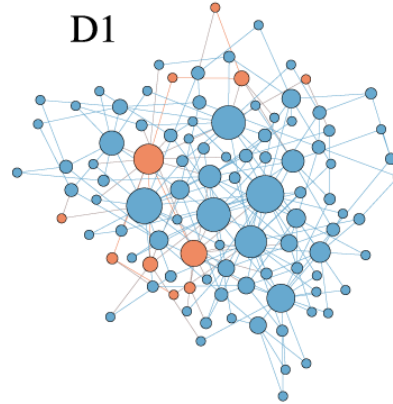
$h = 0.2$



$h = 0.5$

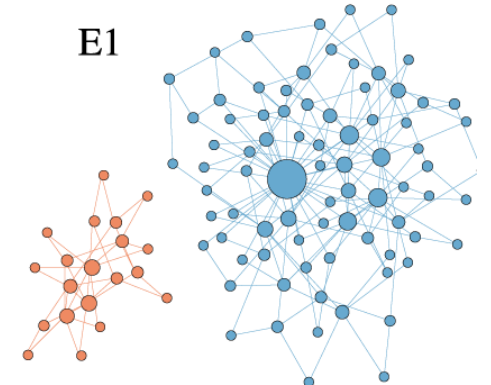


$h = 0.8$



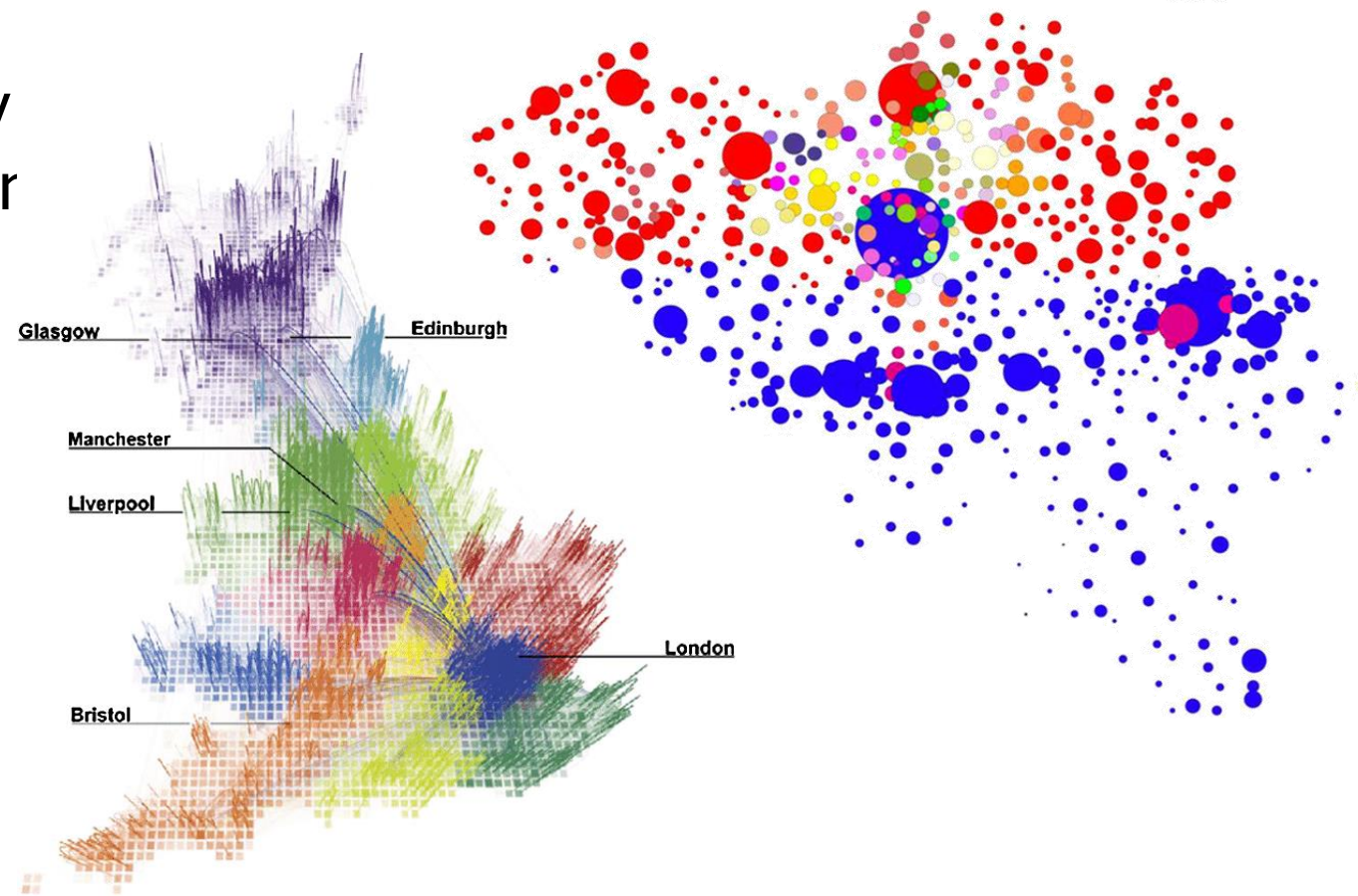
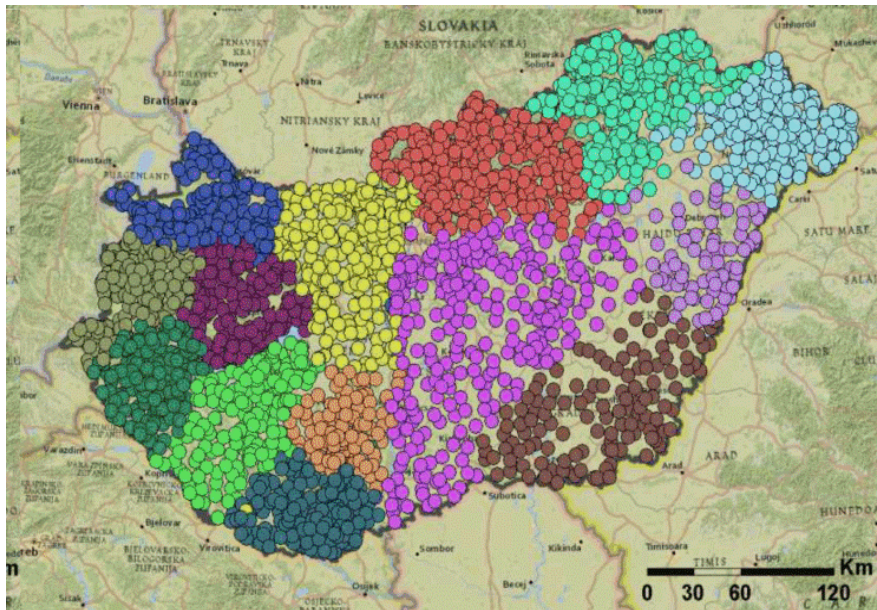
complete homophily

$h = 1$



Spatial modules

- Social networks break down by administrative or cultural region



Residential segregation and social networks

Article | [Open access](#) | Published: 01 August 2022

Social capital I: measurement and associations with economic mobility

[Raj Chetty](#) , [Matthew O. Jackson](#) , [Theresa Kuchler](#) , [Johannes Stroebe](#) , [Nathaniel Hendren](#), [Robert B. Fluegge](#), [Sara Gong](#), [Federico Gonzalez](#), [Armelle Grondin](#), [Matthew Jacob](#), [Drew Johnston](#), [Martin Koenen](#), [Eduardo Laguna-Muggenburg](#), [Florian Mudekereza](#), [Tom Rutter](#), [Nicolaj Thor](#), [Wilbur Townsend](#), [Ruby Zhang](#), [Mike Bailey](#), [Pablo Barberá](#), [Monica Bhole](#) & [Nils Wernerfelt](#)

[Nature](#) **608**, 108–121 (2022) | [Cite this article](#)

182k Accesses | **151** Citations | **2296** Altmetric | [Metrics](#)

Chetty et al. – Social Capital I: Measurement and Associations with Economic Mobility (Nature, Vol. 608 (2022))

Research question: Which form of social capital best predicts upward income mobility?

Data: 21B Facebook friendships (72M users, ages 25–44) – plus Opportunity Atlas, US Census, Civic data

Methods

- Mapped across U.S. counties and ZIP codes
- Regression analysis – EC, Cohesion, Civic Engagement
- Casual tests – Reverse causality, Selection ruled out (race-homogeneous areas), Quasi-experiment confirms causal EC → mobility effect
- Interpretation – Cross-class friendships

Findings

Key results

- Three distinct, weakly correlated dimensions: EC, Cohesion, Civic Engagement
- EC stable over time; varies geographically across U.S.
- EC strongly correlated with upward mobility ($r = 0.65$); others weak or none
- EC–mobility link robust at ZIP level ($r = 0.69$)

Causal evidence

- Reverse causality ruled out (childhood EC)
- Selection ruled out (race-homogeneous areas)
- Quasi-experimental moves confirm causal EC → mobility effect
- +1 EC → +9.8 income percentiles for low-SES children

Interpretation & policy

- EC mediates Cohesion and Civic Engagement effects
- Cross-class friendships drive opportunity
- Fostering cross-SES ties could close 37% of the opportunity gap

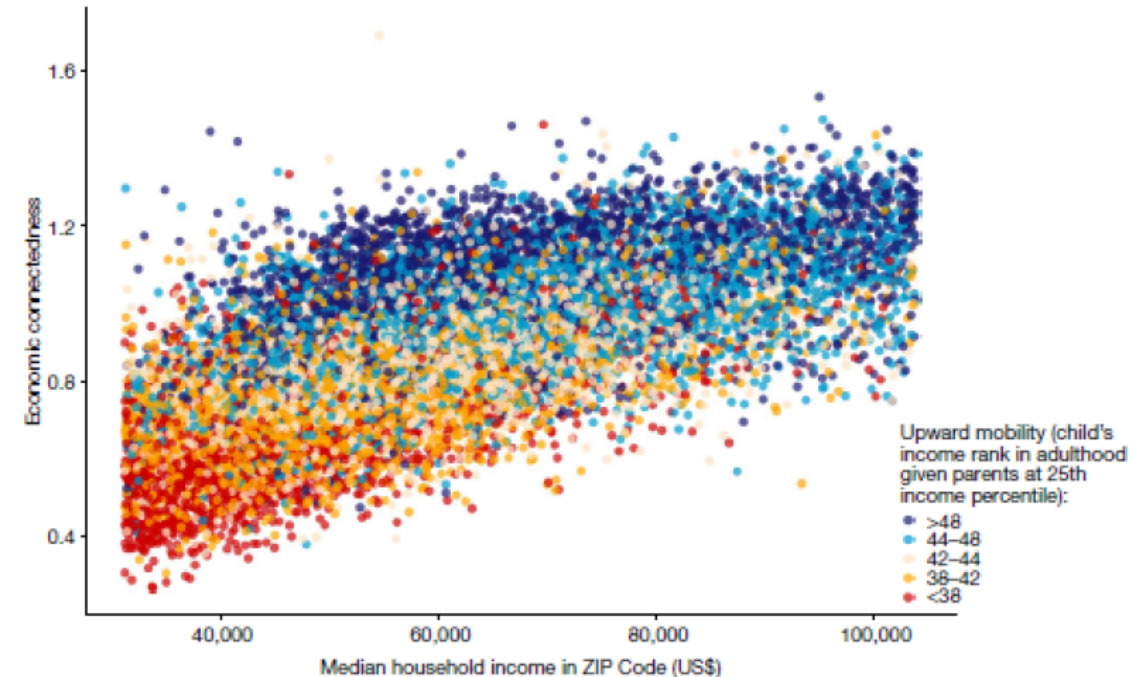


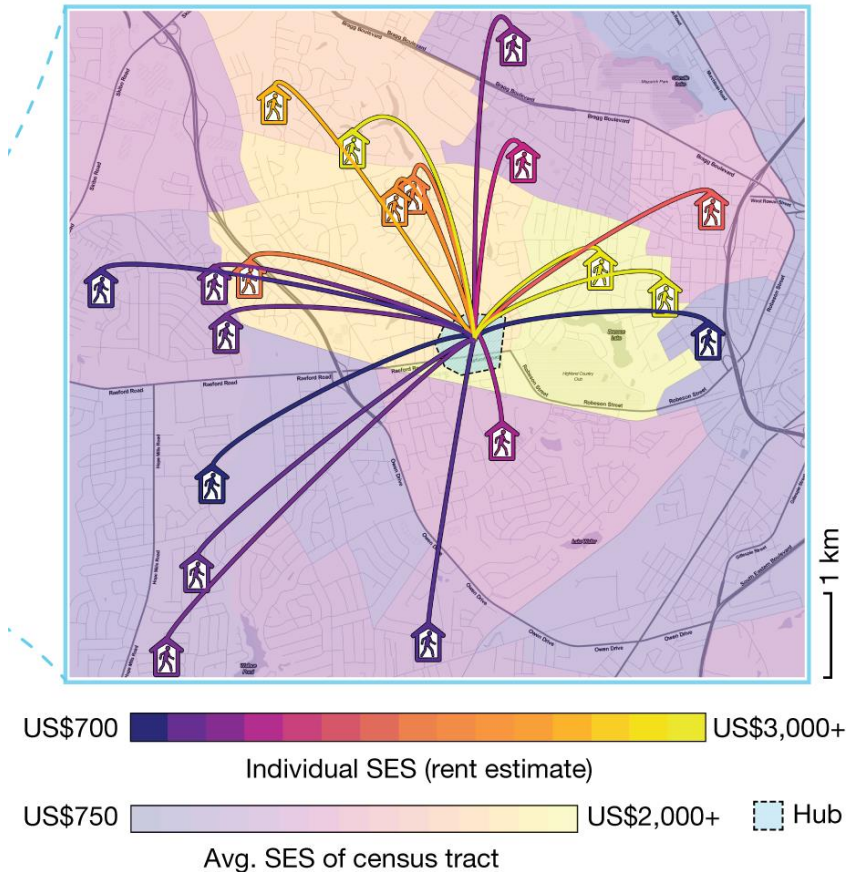
Fig. 6. Upward mobility rises with economic connectedness

Figure: it illustrates the core finding of the study: upward income mobility increases with higher economic connectedness, showing that cross-class friendships — rather than local income levels — drive opportunity.

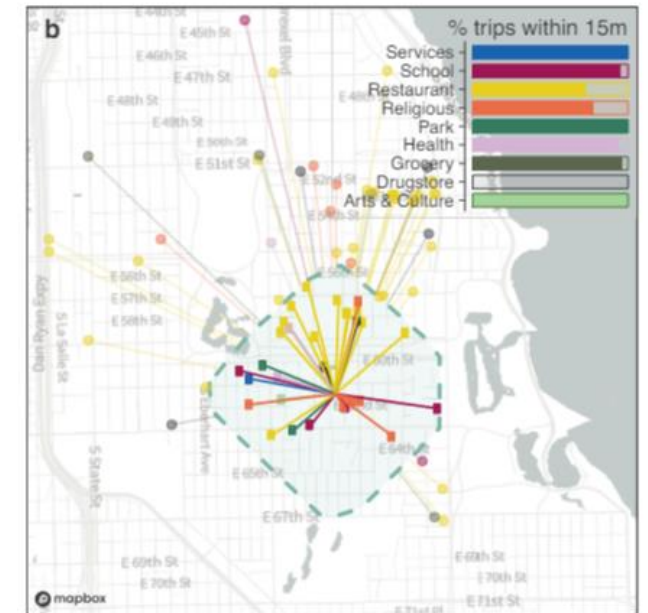
Mobility and experienced segregation



Fan et al. *PNAS Nexus*, 2023.

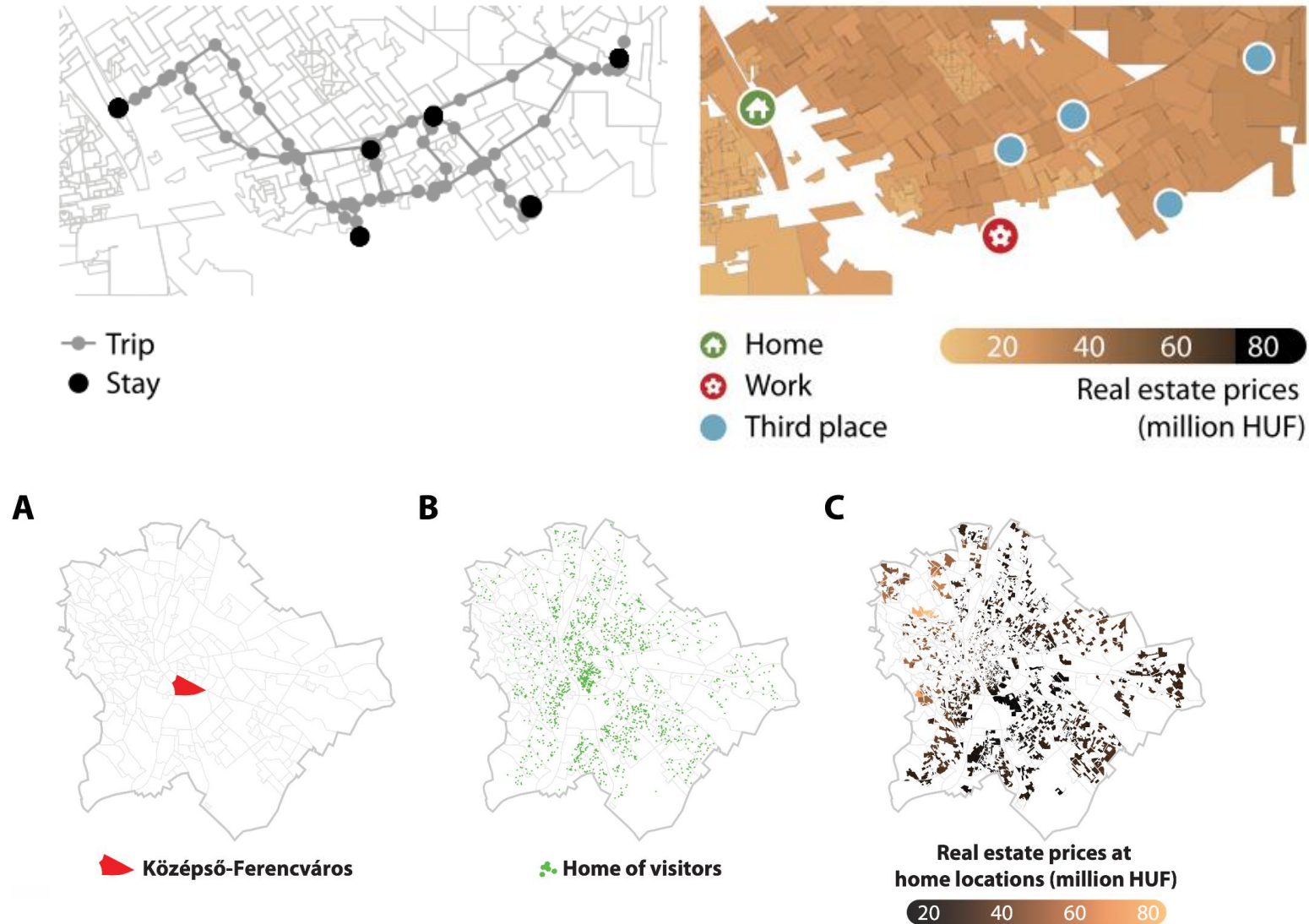


Nilfahorshani et al. *Nature*, 2023.

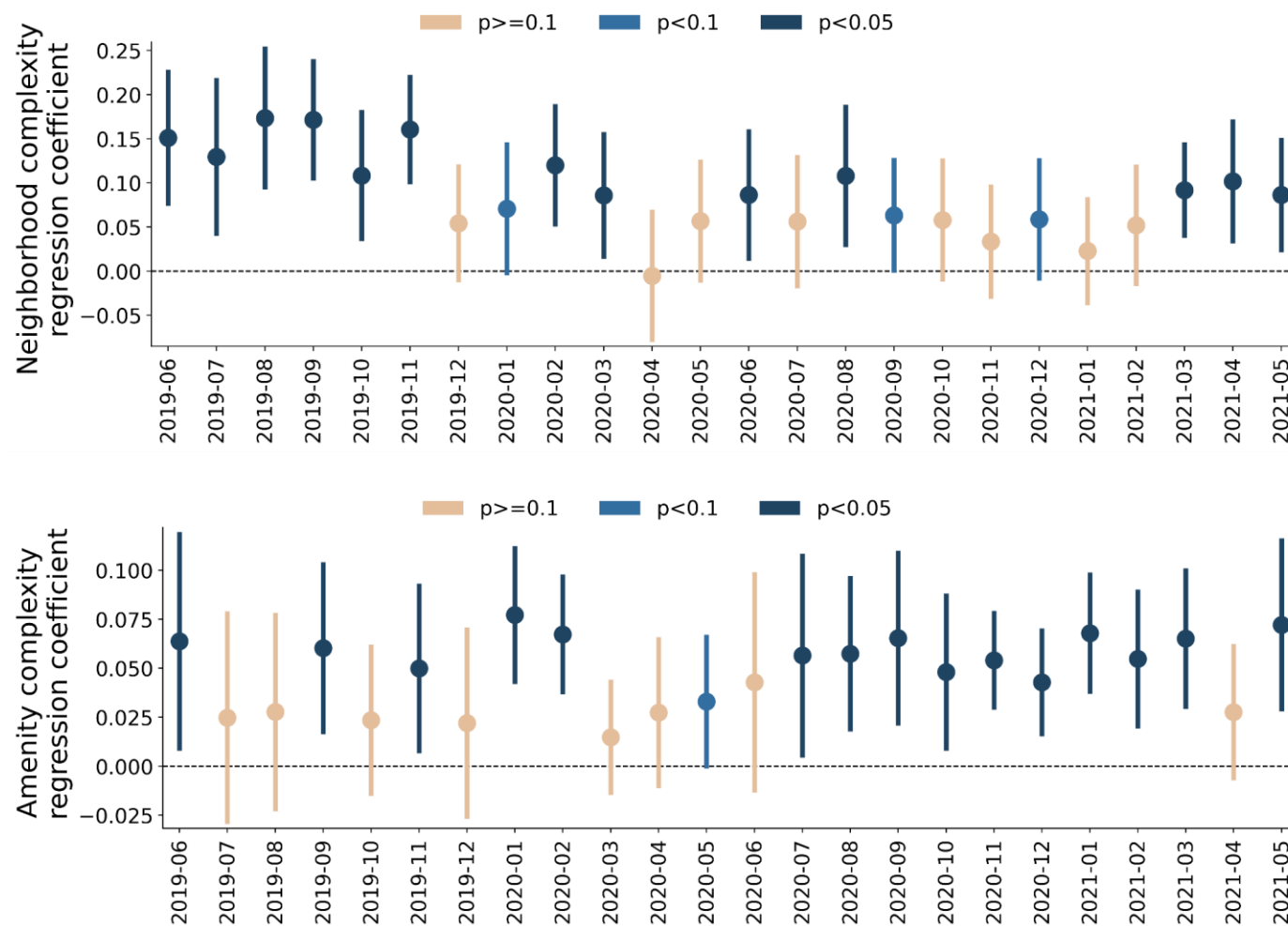


Abbasov et al. *Nature Human Behaviour*, 2024.

Amenity complexity induces social mixing

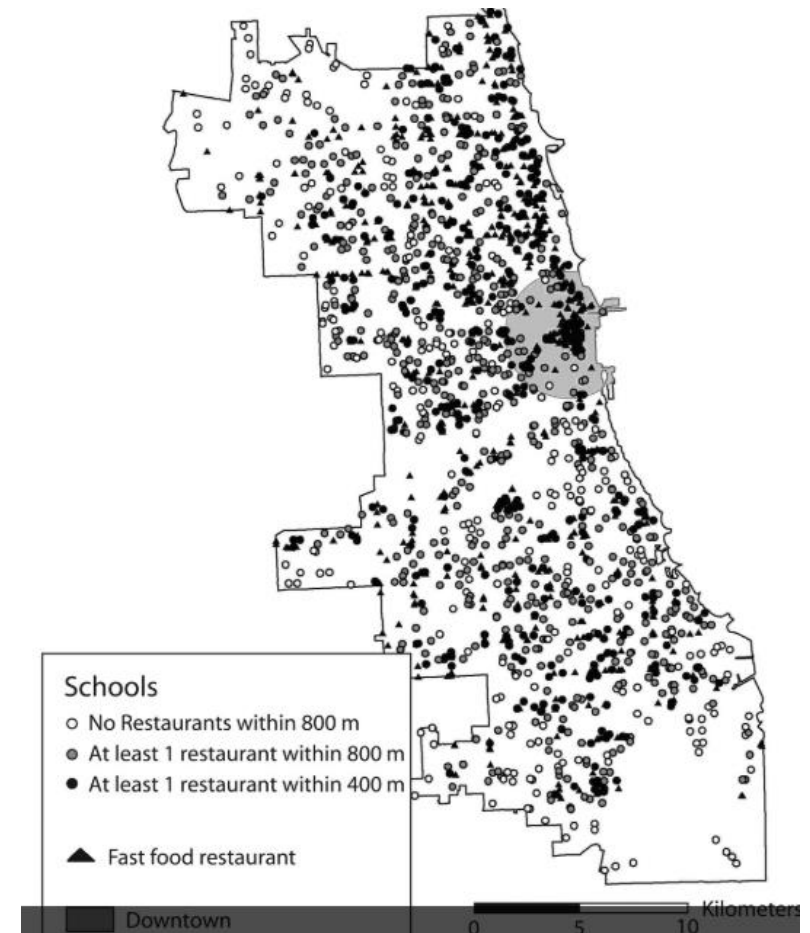


Amenity complexity induces social mixing



Austin, S. B., Melly, S. J., Sanchez, B. N., Patel, A., Buka, S., & Gortmaker, S. L. (2005). Clustering of fast-food restaurants around schools: a novel application of spatial statistics to the study of food environments.

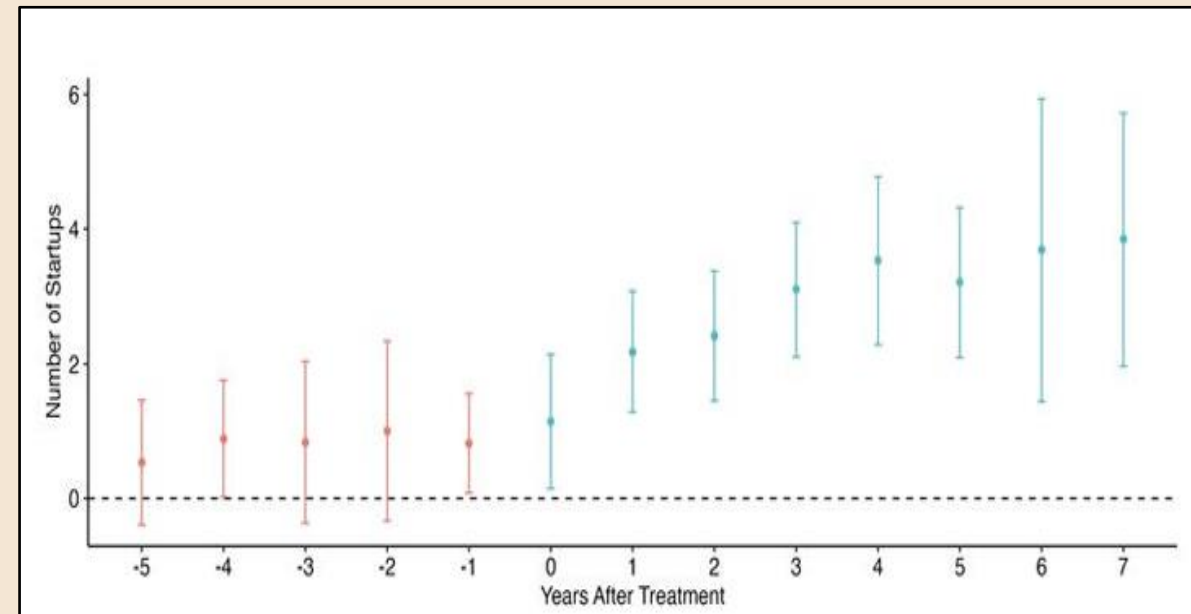
- **Research question:** Do fast-food restaurants concentrate more around schools?
- **Data:** Geocoded 613 restaurants and 1292 schools in Chicago
- **Method:** The authors used the bivariate K function method to quantify the degree of clustering
- **Results:** A significantly greater number of fast-food restaurants are located within a short distance from schools than would be expected if there were no spatial dependence





THIRD PLACES AND NEIGHBORHOOD ENTREPRENEURSHIP: EVIDENCE FROM STARBUCKS CAFÉS

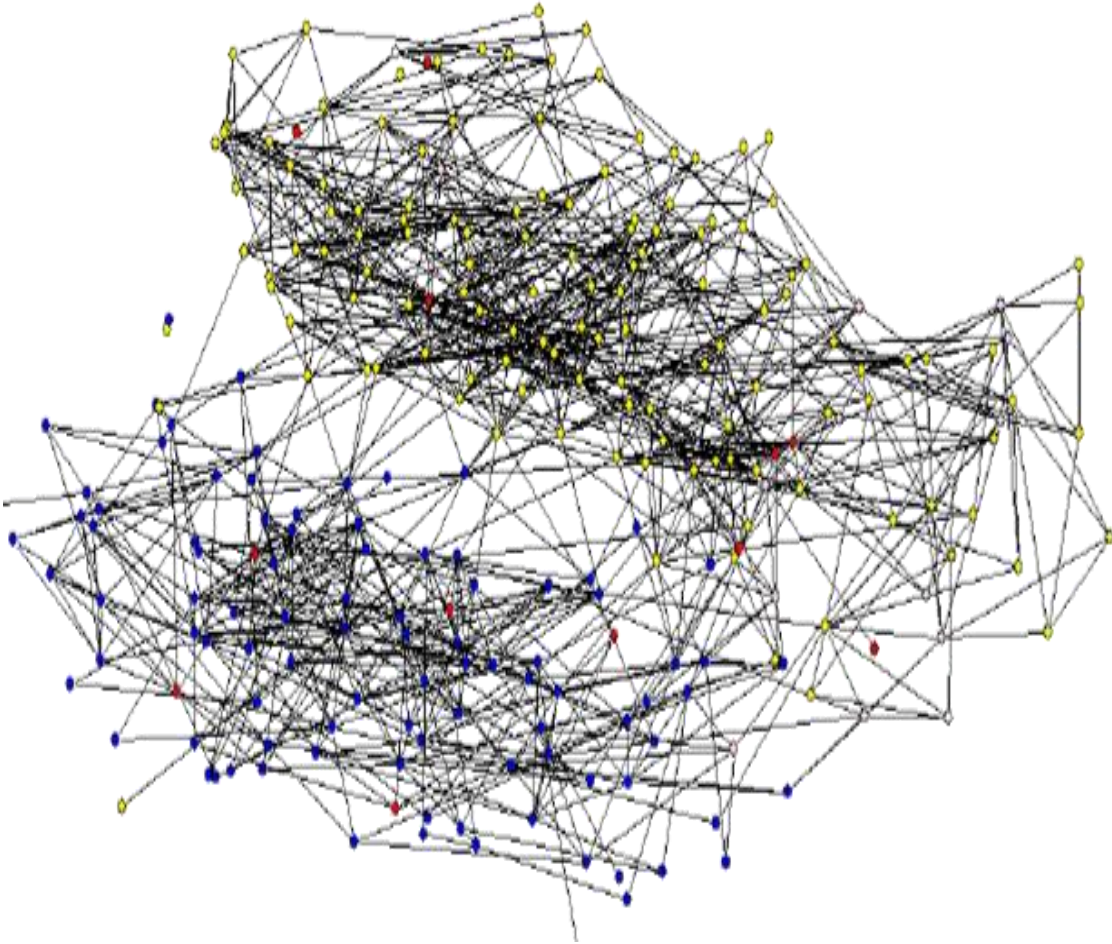
- **Research Question:** Do third places - especially cafés like Starbucks - boost **local entrepreneurial activity** by fostering social interaction and idea exchange?
- **Data:**
 - **main X** = 25,000+ Starbucks openings across the U.S. (2005–2021)
 - **Y** = new business registrations at the ZIP-code level
- **Methodology: Difference-in-differences** method comparing neighborhoods before and after Starbucks entry
- **Findings:**
 - **+4–6% increase** in new business registrations within 2 years of a Starbucks opening
 - Effects are strongest for **knowledge-intensive** and **creative** sectors (tech, design, professional services)
 - Mechanism: cafés act as **“third places”** - informal social hubs where entrepreneurs meet, share info, and build trust



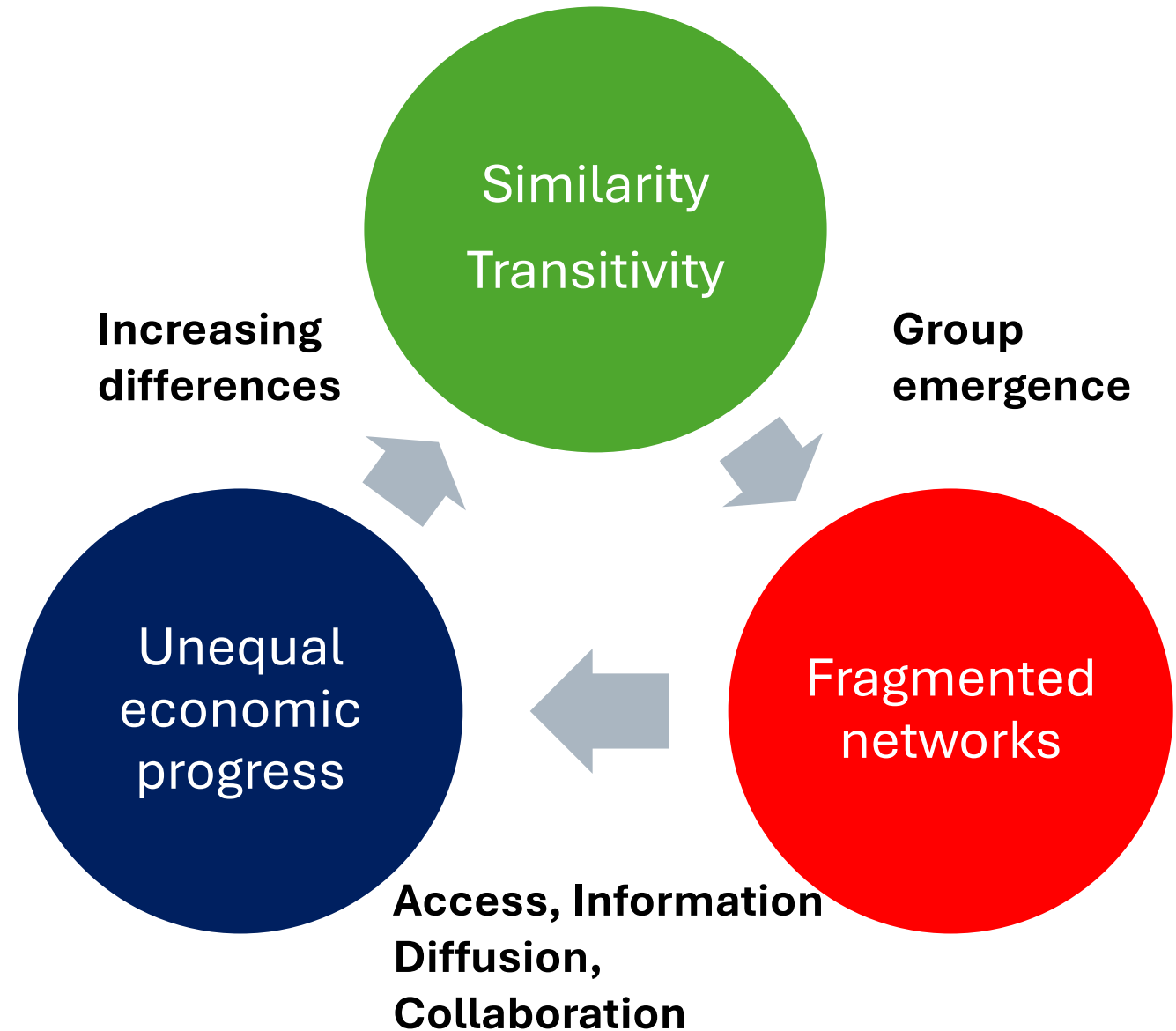
2. Network fragmentation, urban topology and inequalities

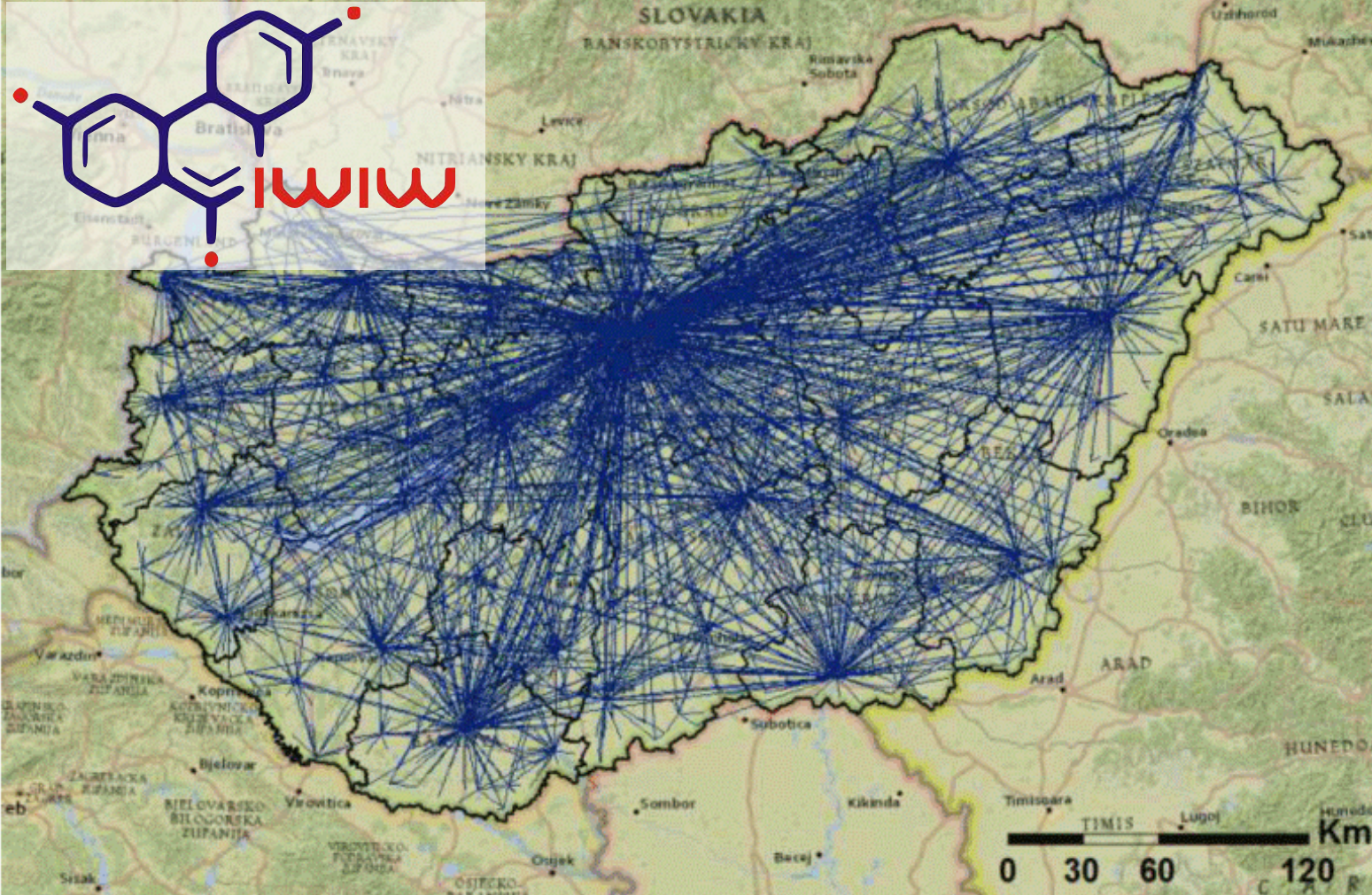
Social networks and inequalities

DiMaggio-Garip (2012) Annual Review of Sociology



Currarini-Pin-Jackson (2009) Econometrica

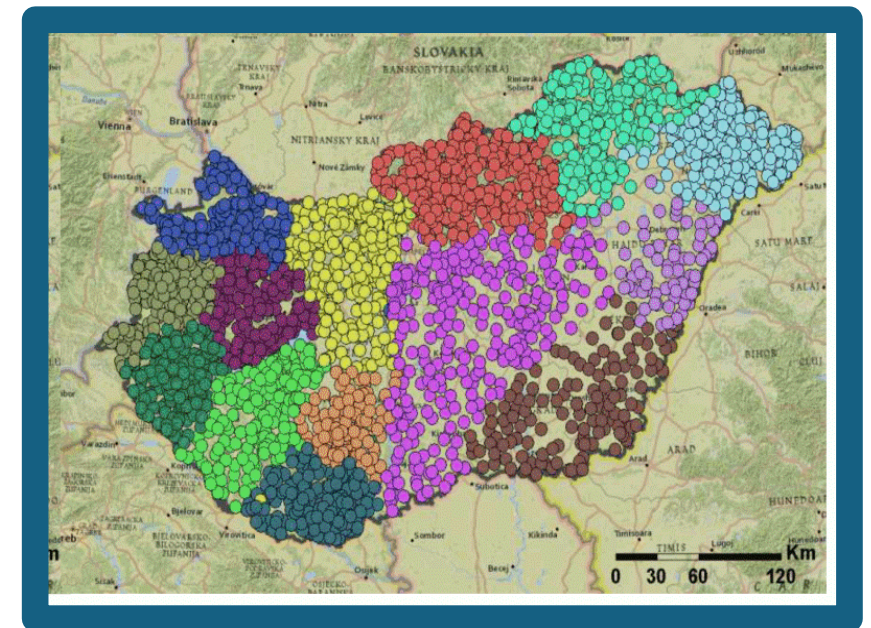




Social media
platform

2.8 M Users

- **Location** (self-reported),
- **Date of registration,**
- **Date of last login,**
- **ID of friends,**
- **ID of invitor**



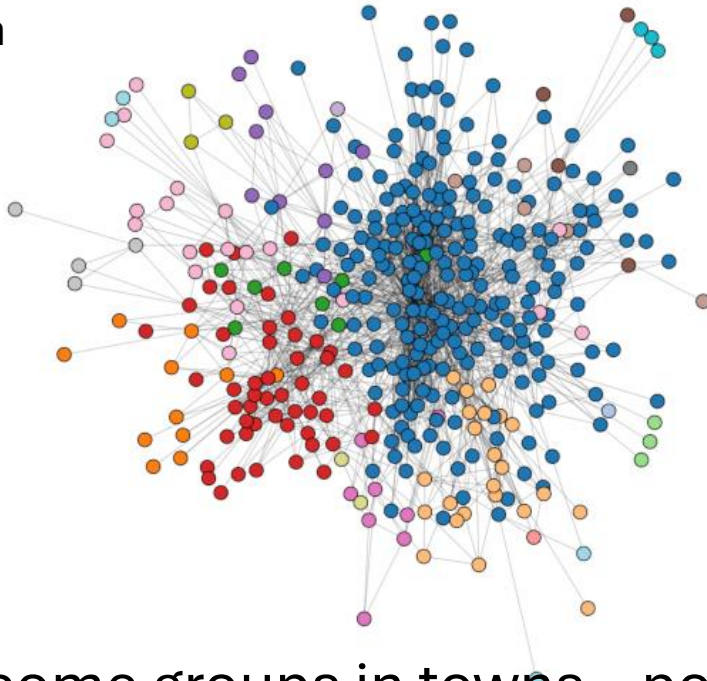
Jakobi and Lengyel (2014) Springer; Lengyel et al. (2015) PLoS ONE

Fragmentation of the network within towns

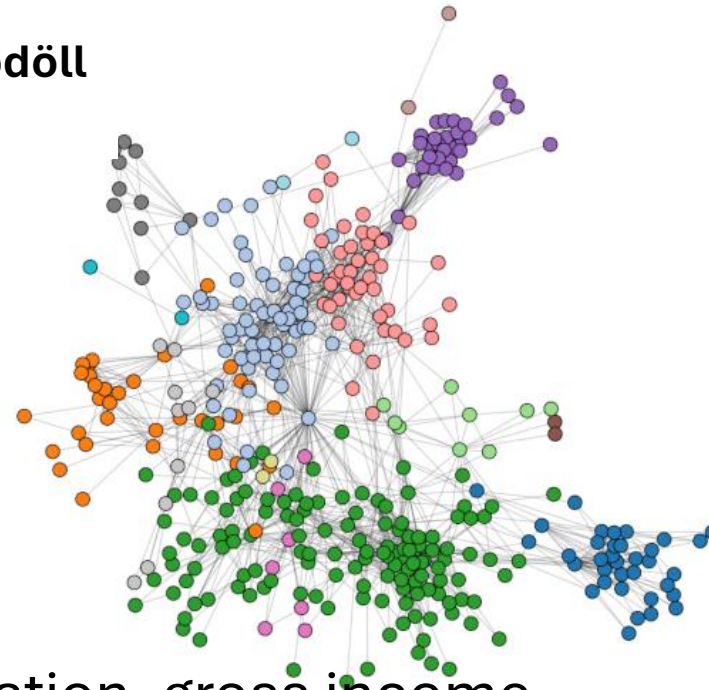
$$Q_i = \sum_{k=1}^{K_i} \left[\frac{L_k^w}{L_i} - \left(\frac{L_k}{L_i} \right)^2 \right]$$

$$F_i = Q_i / Q_i^{max}$$

Ajka

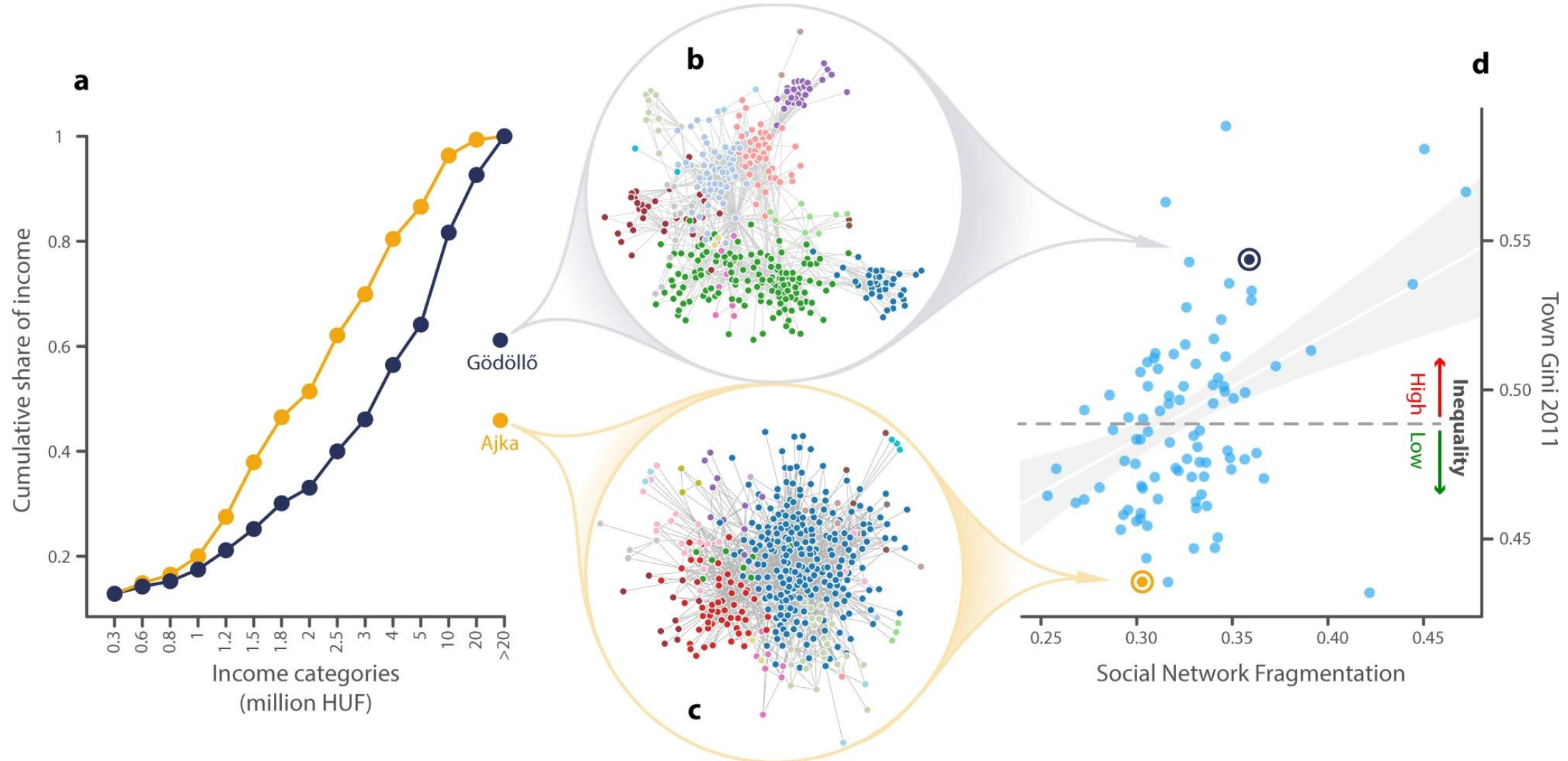


Gödöllő



- TEIR: income groups in towns – population, gross income
- Towns larger than 2500 (except Budapest)
- 2011, 2016

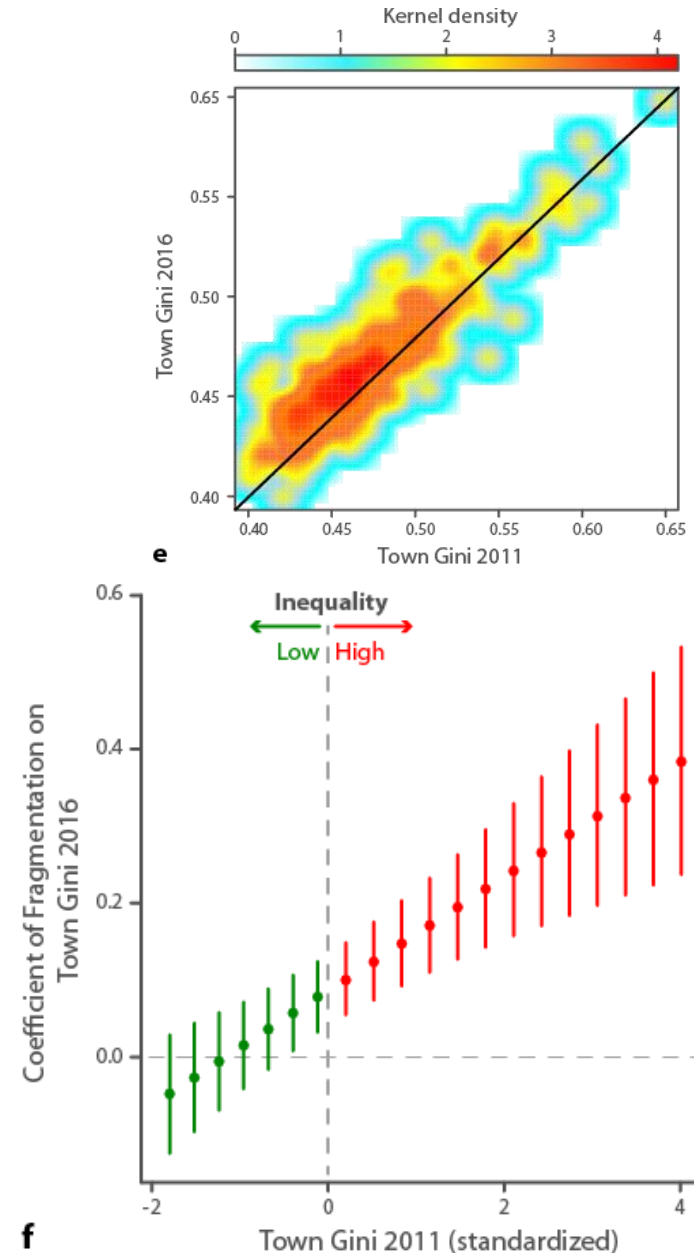
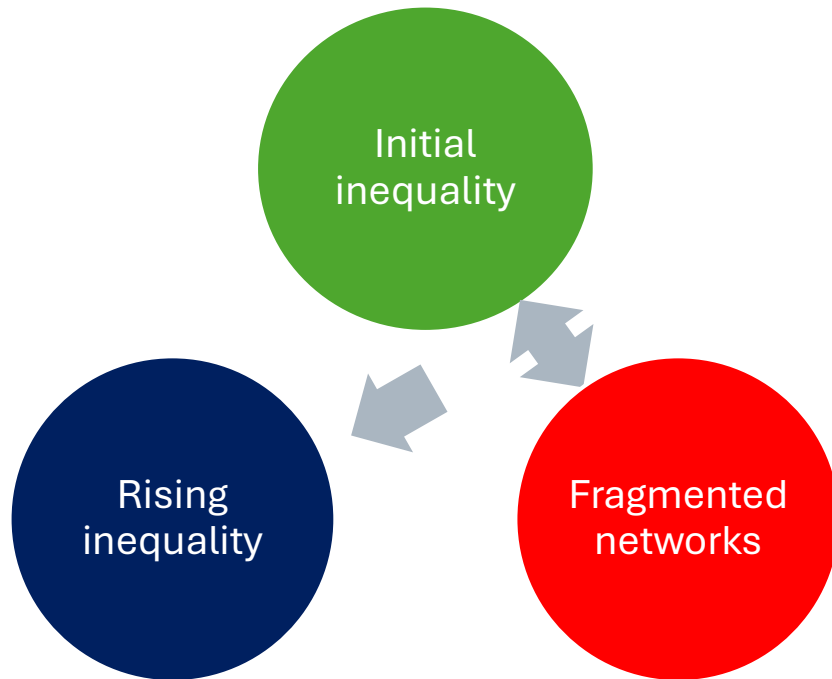
Network fragmentation is correlated with inequalities



Fragmented networks contribute to existing inequalities

$$G_{i,2016} = \alpha \times G_{i,2011} + \beta \times F_i + \gamma \times (G_{i,2011} \times F_i) + Z_{i,2011} + \epsilon$$

$$\frac{\partial G_{i,2016}}{\partial F_i} = \beta + \gamma \times G_{i,2011}$$

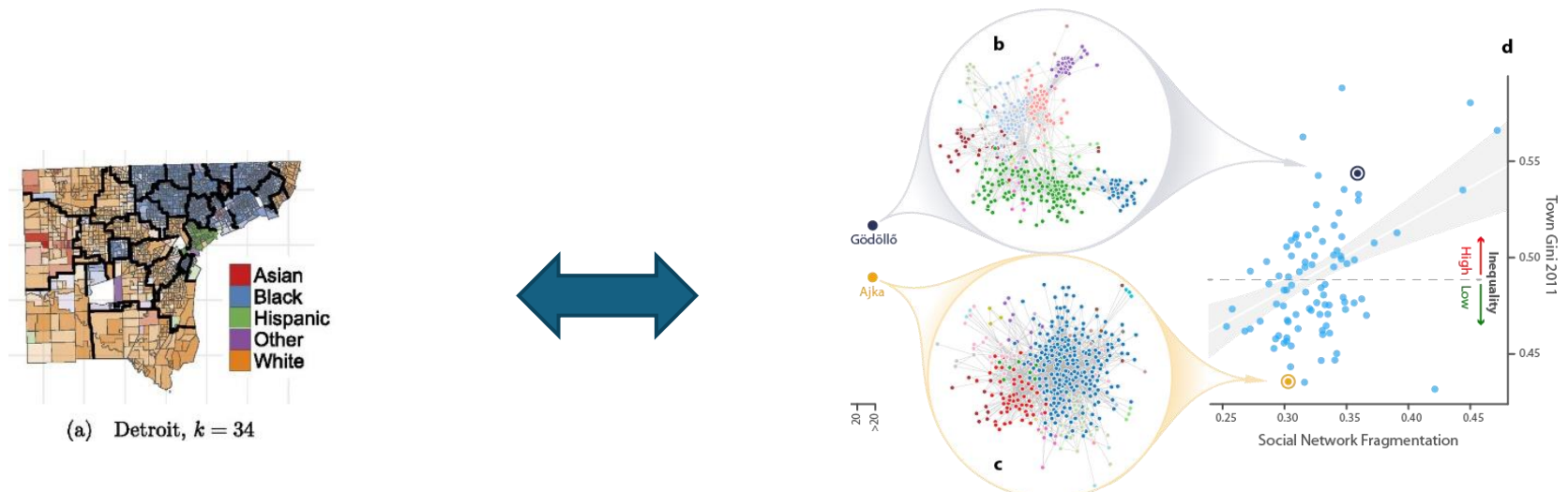


What is the role of urban topology?

A 2SLS approach

$$F_i = \delta + \gamma IV_i + \delta N_i + e_i$$

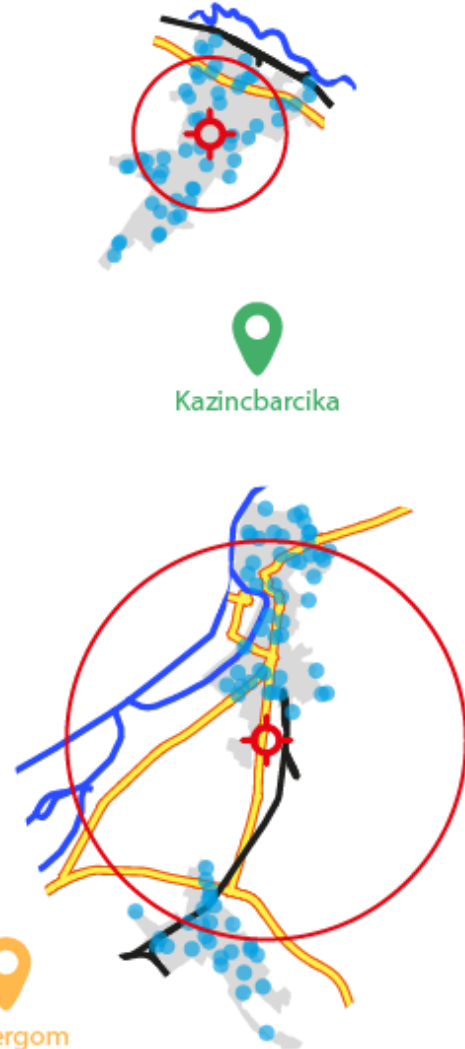
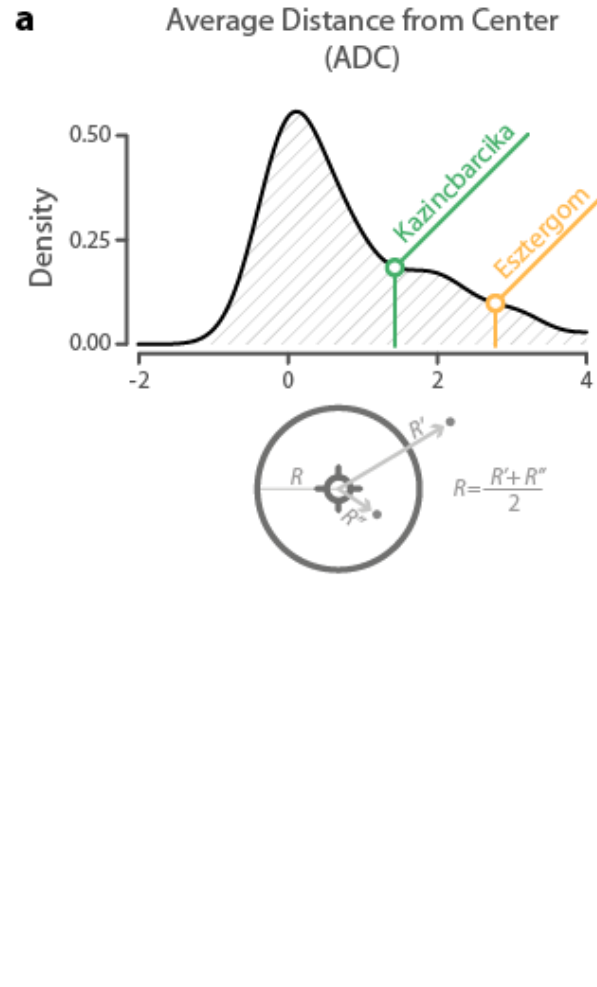
$$G_i = \alpha + \beta_1 \hat{F}_i + \beta_2 X_i + \varphi_k + e_i$$



Urban topology 1: average distance from center

Downtown functions as major hub for social interactions

Urban economics standard models (e.g. von Thünen, Alonso)



$$ADC_i = \frac{\sum_p^P D_{p,c}}{P} / S_i$$

Co-location is important for social interaction and the probability of links decreases as distance grows

-> we expect that fragmentation is higher in towns where ADC is high

Reversed causality is not likely

-> city growth is not likely to be driven by segregated social groups

Urban topology 2: spatial concentration of amenities

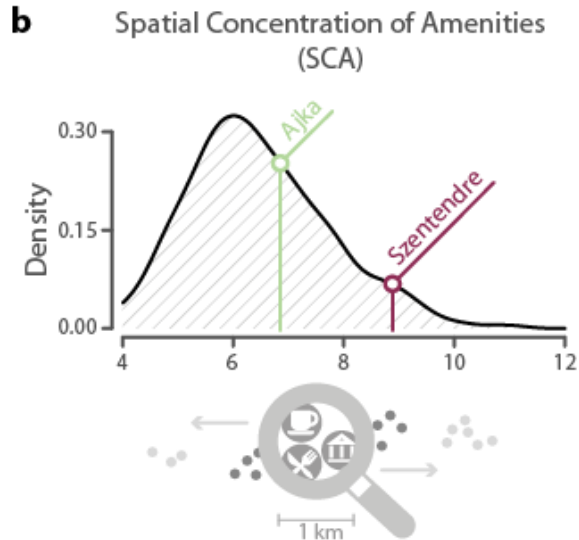
H1: spatial concentration of amenities facilitates segregation of the poor living in the periphery
(Brueckner-Thisse-Zenou, 1999, European Economic Review)

H2: spatial diffuse of amenities provide meeting places in peripheral neighborhoods that facilitates segregation (Zhang-Pryce, 2019, Urban Studies)

$$SCA_i = \frac{\sum_c (p_c \times \log p_c)}{n(c)} / S_i$$

H1 is verified with a positive correlation between the measure and fragmentation.

-> we expect that fragmentation is higher in towns where SCA is high even when controlled for town characteristics.



Urban topology 3: segregation of physical barriers

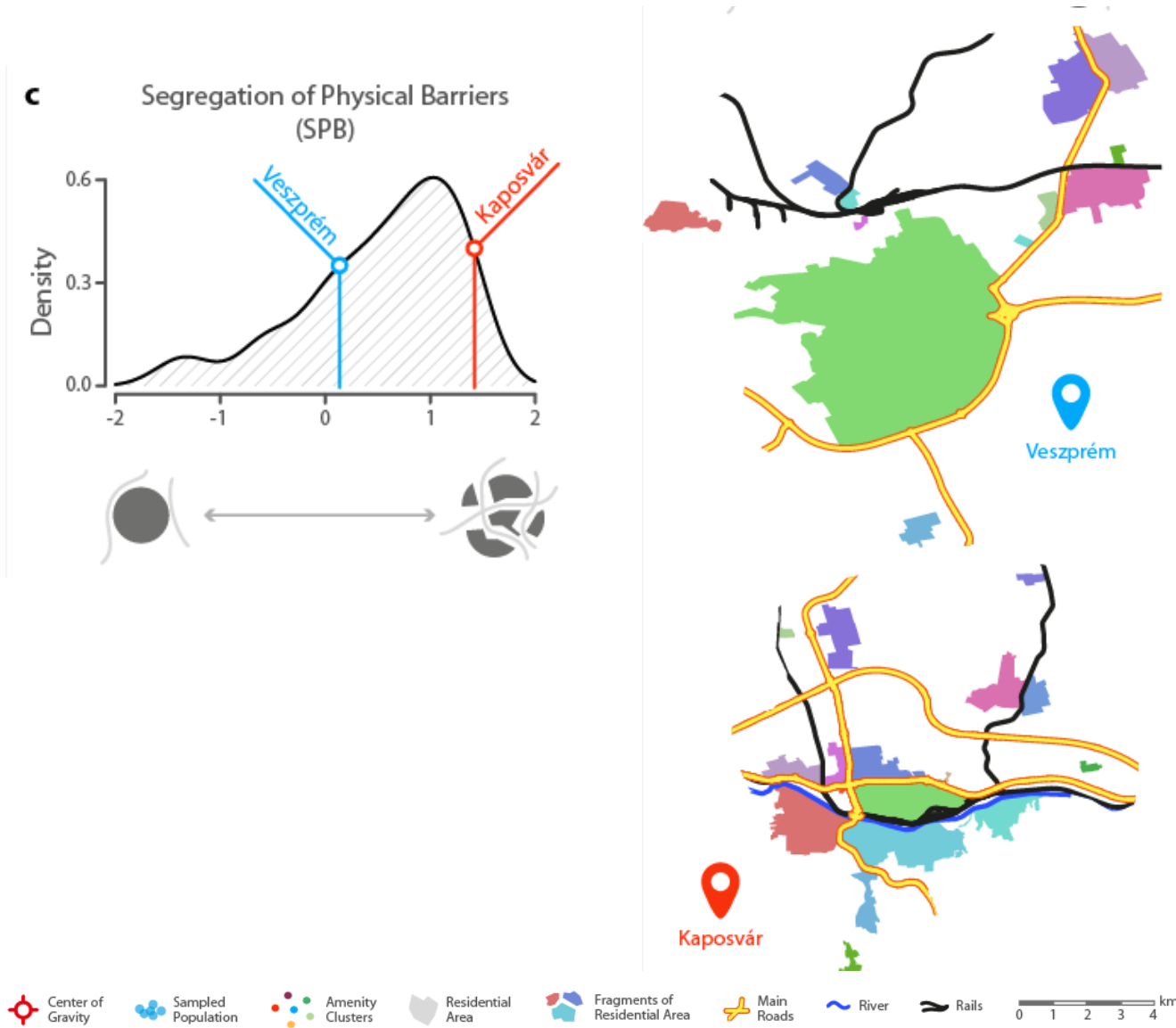
Built and natural barriers facilitate segregation in cities (Cutler and Glaeser, 1997, QJE)

Used to instrument racial segregation effect on inequality (Ananat, 2011, AEJ: Applied Economics)

$$SPB_i = 1 - \sum_a (S_a / S_i)^2$$

Barriers decrease the probability of face-to-face interaction (Lengye et al, 2015 PLOS ONE)
Barriers facilitate sorting into neighborhoods (Benton, 2017, Journal of Urban History)
-> we expect that fragmentation is higher in towns where SPB is high

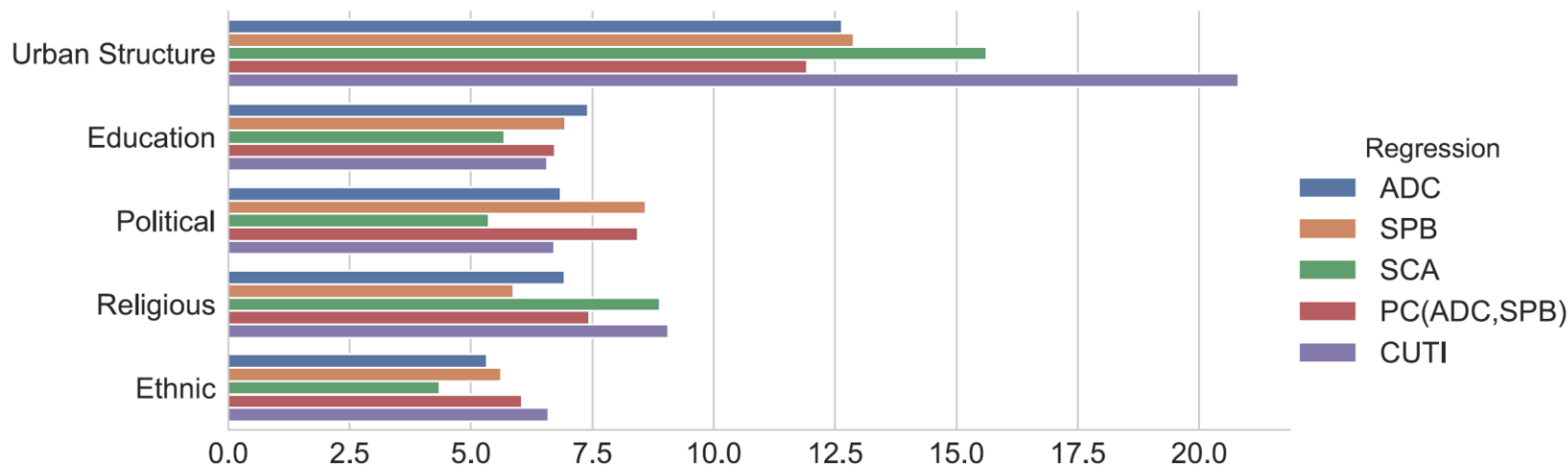
Reversed causality is not likely
-> railroads were not built after the 1940s in Hungary (Körner-Nagy, 2011, Építés-Építészettudomány)



Does urban topology explain network segregation better than other dimensions of social segregation?

Random Forest approach:

- **500 regressions randomly combining the explanatory variables and calculate mean squared error (MSE) from the random sample of these regressions**
- **Randomly vary variables around their mean and recalculate MSE**



1. School inequalities

Coefficient of variance in 6th grade math exam.

2. Political fragmentation

Coefficient of variance in the votes given to Fidesz across voting districts.

3. Religious fragmentation

Entropy of religious distribution.

4. Ethnic fragmentation

Entropy of ethnic distribution.

Does urban topology correlate with network fragmentation? - 1st stage in the 2SLS analysis

$$F_i = \delta + \gamma IV_i + \delta N_i + e_i$$

	Dependent variable: <i>Fragmentation</i> (F_i)		
	(1)	(2)	(3)
<i>ADC</i>	0.091** (0.045)		
<i>SCA</i>		0.110** (0.046)	
<i>SPB</i>			0.168*** (0.044)
User rate	0.367*** (0.045)	0.355*** (0.046)	0.344*** (0.044)
Constant	-0.000 (0.042)	-0.000 (0.042)	-0.000 (0.042)
Observations	473	473	473
R ²	0.167	0.170	0.185
Adjusted R ²	0.163	0.166	0.181
Residual Std. Error (df = 470)	0.915	0.913	0.905
F Statistic (df = 2; 470)	47.082***	48.085***	53.259***

Note:

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Urban topology <-> network fragmentation <-> inequality

2nd stage in the 2SLS analysis

$$G_i = \alpha + \beta_1 \hat{F}_i + \beta_2 X_i + \varphi_k + e_i$$

Control variables:

- Level of FDI
- Change of FDI
- Unemployment rate
- Population density
- Distance to the closest border

Dependent variable: Gini Coefficient ($G_{i,2016}$)			
	<i>Instrumental Variable</i>		
	<i>ADC</i>	<i>SCA</i>	<i>SPB</i>
Estimated Fragmentation ($\hat{F}_i$)	0.408*** (0.153)	0.533*** (0.146)	0.288** (0.138)
Population density	-0.092* (0.055)	-0.118** (0.052)	-0.067 (0.053)
Distance to border	-0.243*** (0.059)	-0.231*** (0.062)	-0.254*** (0.058)
Constant	-0.386 (0.369)	-0.392 (0.330)	-0.380 (0.372)
County FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
First Stage F -test	22.290***	24.009***	26.754***
Wu-Hausman test	1.107	3.729	0.011
Sargan test	0.051	5.349*	1.400
Observations	473	473	473
R ²	0.231	0.192	0.245
Adjusted R ²	0.186	0.145	0.200
Res.St.Err. (df = 446)	0.902	0.924	0.894

Note:

*p<0.1; **p<0.05; ***p<0.01

3. Social media and urban inequalities

Sándor Juhász

Income-related spatial concentration of individual social capital in cities

Ádám J Kovács

Agglomeration and Social Networks Lendület Research Group, ELKH KRTK, Budapest Tóth Kálmán u. 4

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Abstract

Social connections that span across diverse urban neighborhoods can support prosperity by mobilizing social capital. However, there is limited evidence on the spatial structure of individual social capital inside cities. This paper demonstrates that social capital measured by online social connections is spatially more concentrated for residents of lower-income neighborhoods than for residents of higher-income neighborhoods. We map the micro-geography of individual online social networks in the 50 largest metropolitan areas of the United States using a large-scale geolocalized Twitter dataset. We analyze the spatial dimension of individual social capital by the share of friends, closed triangles, and share of supported ties within circles of short distance radii (1, 3, 5, and 10 km) around users' home location. We compare residents from below-median income neighborhoods with above-median income neighborhoods, and find that users living in relatively poorer neighborhoods have a significantly higher share of connections in close proximity. Moreover, their network is more cohesive and supported within a short distance from their home. These patterns prevail across the 50 largest US metropolitan areas with only a few exceptions. The found disparities in the micro-geographic concentration of social capital can feed segregation and income inequality within cities constraining social circles of low-income residents.

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scientific reports

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Universal patterns of long-distance commuting and social assortativity in cities

Eszter Bokányi^{1,2,3}, Sándor Juhász^{1,2}, Márton Karsai^{1,4} & Balázs Lengyel^{1,2}

Millions commute to work every day in cities and interact with colleagues, partners, friends, and strangers. Commuting facilitates the mixing of people from distant and diverse neighborhoods, but whether this has an imprint on social inclusion or instead, connections remain assortative is less explored. In this paper, we aim to better understand income sorting in social networks inside cities and investigate how commuting distance conditions the online social ties of Twitter users in the 50 largest metropolitan areas of the United States. An above-median commuting distance in cities is linked to more diverse individual networks, moreover, we find that longer commutes are associated with a nearly uniform, moderate reduction of overall social tie assortativity across all cities. This suggests a universal relation between long-distance commutes and the integration of social networks. Our results inform policy that facilitating access across distant neighborhoods can advance the social inclusion of low-income groups.

Cities are champions of diversity^{1–3}. Complex interaction networks of individuals in urban areas enabled by population density, co-location, and easy access together made cities the global engines of technological and economic progress^{4–7}. However, cities are also known for high levels of segregation^{8–10} where disparate neighborhoods are separated from each other in the urban space^{11–13}. Furthermore, spatial segregation by income also fragments social networks, which can hinder progress and can deepen inequalities^{14–20}. Given the importance of this problem, a growing community has investigated the patterns of mobility in cities to better understand mixing potentials across disparate and diverse neighborhoods^{21–24}, which may increase economic prosperity²⁵. Yet, less is known whether mobility mixing has any imprint on the social connections of people.

Commuting covers a large share of urban mobility²⁶ and by connecting home with work locations, the places where people spend most of their time, it plays an important role in the spatial formation of social connections^{27–29}. Since aggregated social networks form spatially bounded communities across neighborhoods³⁰, the further one commutes, the higher the likelihood that commuting-related social connections will introduce diversity in the egocentric network of the commuter^{31,32}. Due to spatial segregation, economically disparate neighborhoods tend to be far from each other³³, thus long commutes are more likely to link places with different social status^{34,35}. Nevertheless, it is not trivial that long commutes should facilitate social inclusion, because social interactions might remain assortative even at places far from home^{32,33,37}. Meanwhile, the time to develop new social connections is especially limited for low-income workers who travel to work during rush hours^{38–41}.

The spatial distribution of high versus low-income households determines the length of travel that can bridge disparate neighborhoods. Since the scale of socio-economic isolation greatly varies across cities⁴², one may expect that the mobility of people also enables a different degree of social mixture. However, the assortativity of urban mobility is a universal feature across cities: individuals have been recently reported to visit locations that are similar to their home neighborhoods^{43,44–46}. Yet, how assortativity of commuting and social networks are related and how this relation is modified by the length of commute in cities is still largely uncovered.

In this paper, we aim to better understand how mixing in urban social networks is facilitated by commuting. To answer this question, we use a unique dataset on 348,850 Twitter users living in the 50 largest metropolitan areas of the US and track their home and work locations as well as their mutual followership ties on the platform, which from now on, we call the social network of users. We project these social networks in the urban space and attribute users with an average income based on their home locations on an income map extracted from census

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Urban highways are barriers to social ties

Luca Maria Aiello^{1,2,3,4}, Anastassia Vybormova^{1,2}, Sándor Juhász^{1,4,5}, Michael Szell^{1,6,7,8}, and Eszter Bokányi⁹

Affiliations are included on p. 9.

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Urban highways are common, especially in the United States, making cities more car-centric. They promise the annihilation of distance but obstruct pedestrian mobility, thus playing a key role in limiting social interactions locally. Although this limiting role is widely acknowledged in urban studies, the quantitative relationship between urban highways and social ties is barely tested. Here, we define a Barrier Score that relates massive, geolocalized online social network data to highways in the 50 largest US cities. At the granularity of individual social ties, we show that urban highways are associated with decreased social connectivity. This barrier effect is especially strong for short distances and consistent with historical cases of highways that were built to purposefully disrupt or isolate Black neighborhoods. By combining spatial infrastructure with social tie data, our method adds a dimension to demographic studies of social segregation. Our study can inform reparative planning for an evidence-based reduction of spatial inequality, and more generally, support a better integration of the social fabric in urban planning.

social network | segregation | highways | barrier effect

Cities are hubs of concentrated social capital that can foster diversity and innovation (1, 2). However, this potential is threatened by spatial fragmentation through built infrastructure that can divide neighborhoods (3, 4), exacerbate inequalities (5, 6), and contribute to segregation (7). Among various types of barriers fragmenting urban areas, roads designed for motorized traffic are the most ubiquitous, especially highways (4, 8, 9). Since the 1960s, urban planners have theorized that high-traffic roads reduce opportunities for creating and maintaining social ties across divided neighborhoods (10), thus undermining the social cohesion essential for the development of thriving communities. This premise lies at the core of contemporary urban planning research and interventions (11, 12) that strive to meet the UN's sustainable development goal of "making cities and human settlements inclusive, safe, resilient, and sustainable" (13).

Despite its significance in urban planning theories, the association between high-traffic roads and reduced social connectivity has never been measured empirically, with the notable exception of a few small-scale, survey-based studies (14, 15). Previous quantitative research in this area, constrained by the scarcity of georeferenced social network data (16, 17), has focused instead on measuring socioeconomic segregation in cities. This goal has been achieved either by using static demographic data (7, 18) or, more recently, through mobility data (19–21), with only sporadic attempts to link segregation to urban barriers (6, 22, 23). While highly valuable, such previous research could not explicitly consider social ties. However, providing an explicit, quantitative estimation of the barrier effect of different roads in curbing social ties is crucial for guiding evidence-based plans of restorative urban interventions and for prioritizing them according to their estimated benefits (24).

To fill this gap, we introduce a method to systematically quantify the association between highways and social ties at multiple scales, ranging from individual highway segments to entire metropolitan areas. We focus on the network of urban highways in the United States. This highway network offers a compelling subject for the study of barrier effects: With a cost of at least 1.4 trillion USD (25), US highways were built to bridge city centers and newly created suburbs; simultaneously, they displaced an estimated 1 million people from their neighborhoods and today pose hard-to-cross physical barriers to pedestrians and cyclists (4, 26).

Onto this network of urban highways within the 50 largest metropolitan areas in the United States, we overlay a massive geolocalized social network of ties between individuals who follow each other on Twitter (27). We compute a Barrier Score which quantifies the reduction in the number of social ties crossing highways, comparing the empirical crossings with a null model that makes ties oblivious to highways. The distribution

Significance

Highways are physical barriers that cut opportunities for social connections, but the magnitude of this effect has not been quantified. Such quantitative evidence would enable policy-makers to prioritize interventions that reconnect urban communities—an urgent need in many US cities. We relate urban highways in the 50 largest US cities with massive, geolocalized online social network data to quantify the decrease in social connectivity associated with highways. We find that this barrier effect is strong in all 50 cities, and particularly prominent over shorter distances. We also confirm this effect for highways that are historically associated with racial segregation. Our research demonstrates with high granularity the long-lasting impact of decades-old infrastructure on society and provides tools for evidence-based remedies.

Author contributions: L.M.A., A.V., S.J., M.S., and E.B. designed research; L.M.A., A.V., S.J., M.S., and E.B. performed research; L.M.A., A.V., and S.J. contributed new reagents/analytic tools; L.M.A., A.V., S.J., M.S., and E.B. analyzed data; and L.M.A., A.V., S.J., M.S., and E.B. wrote the paper.

The authors declare no competing interest.

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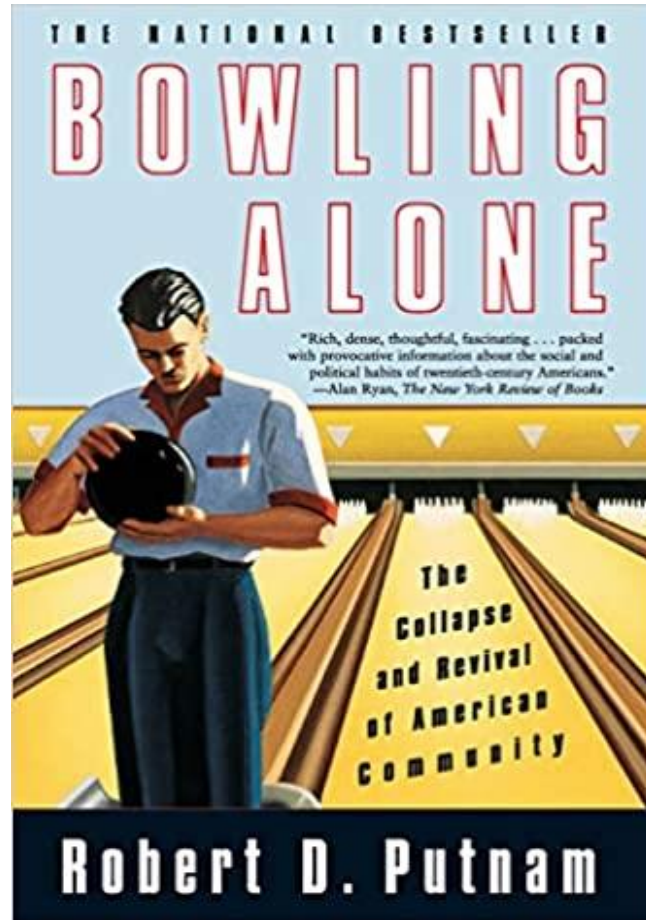
[†]To whom correspondence may be addressed. Email: kua@bu.dk.

This article contains supporting information online at <https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2408937122/-/DCSupplemental>.

Published March 4, 2025.

3.1) Income-related spatial concentration of individual social capital in cities

Social capital, social networks and geographic constraints



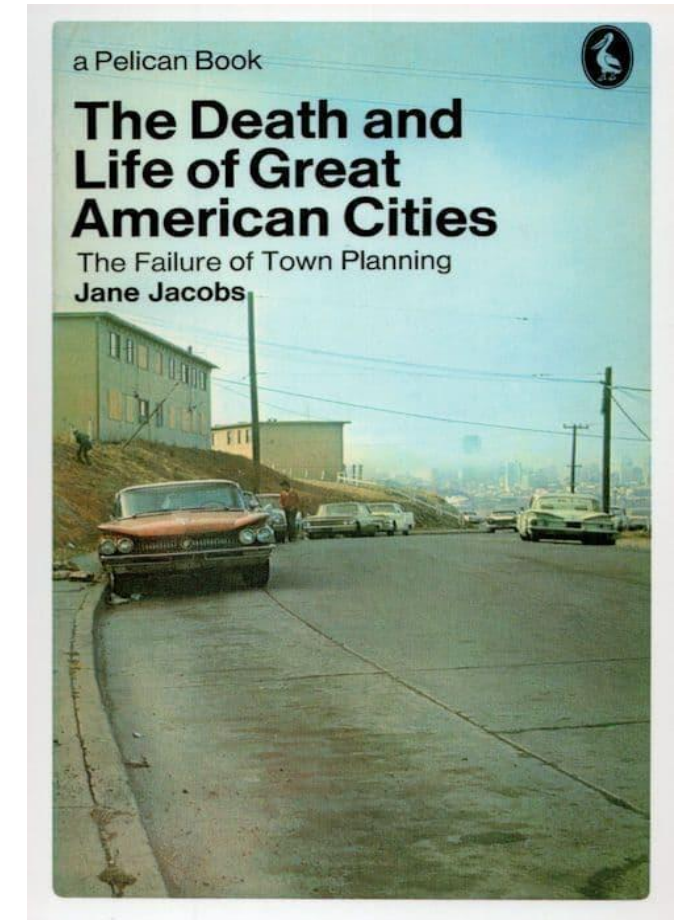
Social Capital in the Creation of Human Capital¹

James S. Coleman
University of Chicago

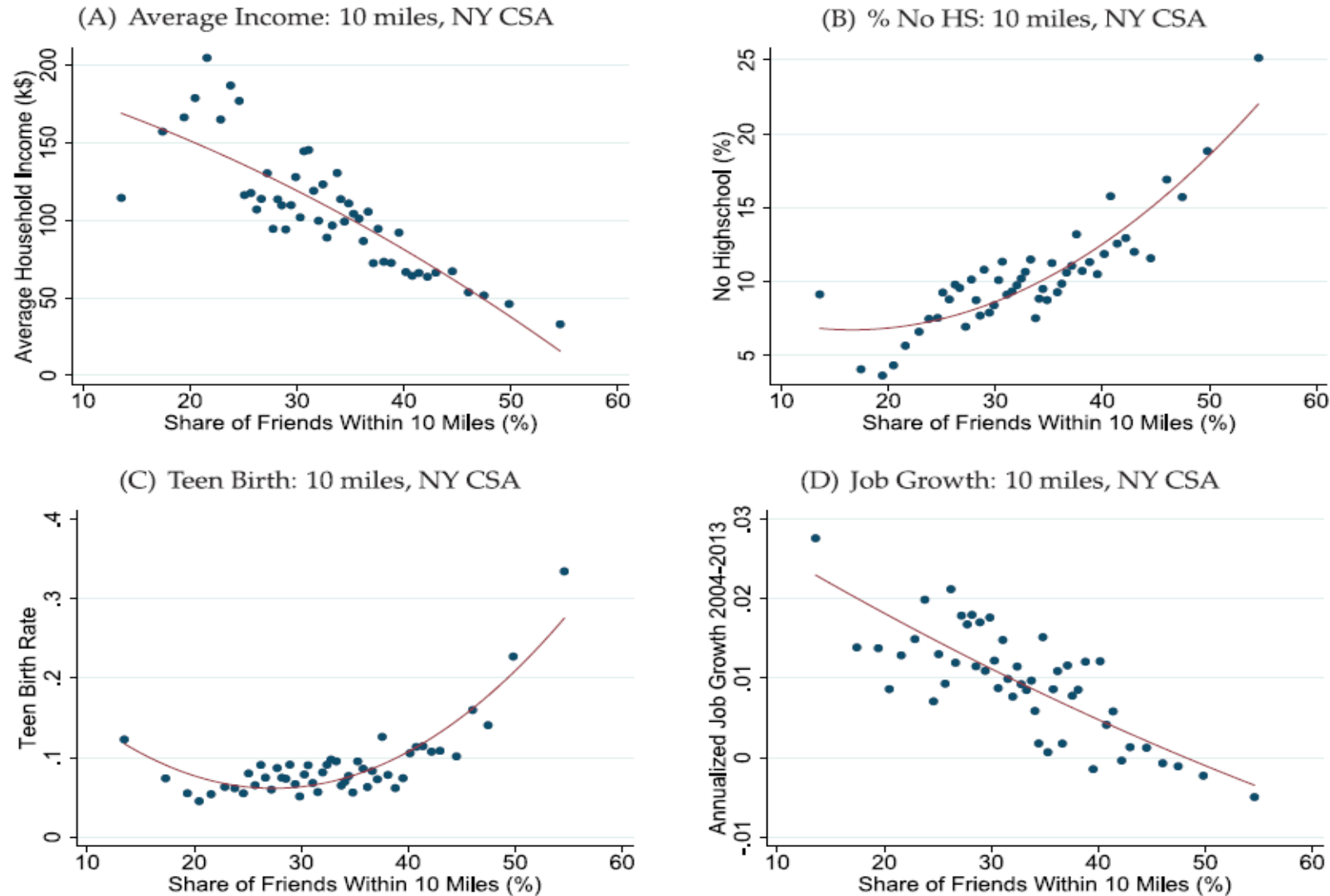
In this paper, the concept of social capital is introduced and illustrated, its forms are described, the social structural conditions under which it arises are examined, and it is used in an analysis of dropouts from high school. Use of the concept of social capital is part of a general theoretical strategy discussed in the paper: taking rational action as a starting point but rejecting the extreme individualistic premises that often accompany it. The conception of social capital as a resource for action is one way of introducing social structure into the rational action paradigm. Three forms of social capital are examined: obligations and expectations, information channels, and social norms. The role of closure in the social structure in facilitating the first and third of these forms of social capital is described. An analysis of the effect of the lack of social capital available to high school sophomores on dropping out of school before graduation is carried out. The effect of social capital within the family and in the community outside the family is examined.

There are two broad intellectual streams in the description and explanation of social action. One, characteristic of the work of most sociologists, sees the actor as socialized and action as governed by social norms, rules, and obligations. The principal virtues of this intellectual stream lie in its ability to describe action in social context and to explain the way action is shaped, constrained, and redirected by the social context.

The other intellectual stream, characteristic of the work of most economists, sees the actor as having goals independently arrived at, as acting independently, and as wholly self-interested. Its principal virtue lies in



Space and socio-economic dimensions of social ties



Micro-geography of individual social capital in cities correlate with the income of residence?

Data



Sándor Juhász
@sador_juhasz

Can we explain how relatedness of technologies evolves over time? Our new paper in Papers in Regional Science with @TBroekel and @rboschma identifies the drivers behind the widely used concept of #relatedness rsaconnect.onlinelibrary.wiley.com/doi/full/10.1111/pirs.12345 Quick thread 1/3



Explaining the dynamics of relatedness: The role of co-lo...
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Sándor Juhász
@sador_juhasz

Researcher at NETI Lab and ANET Lab | Interested in data magic, networks, innovation and (economic) geography

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Sándor Juhász
@sador_juhasz



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Twitter Web App

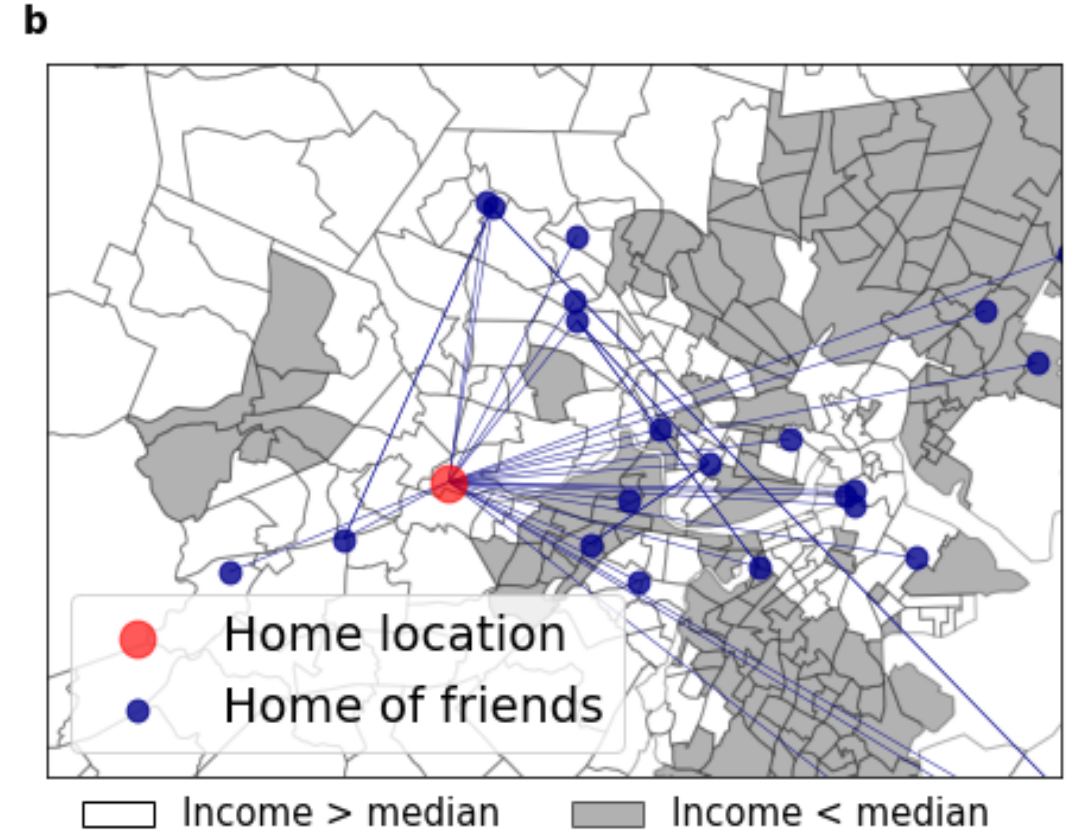
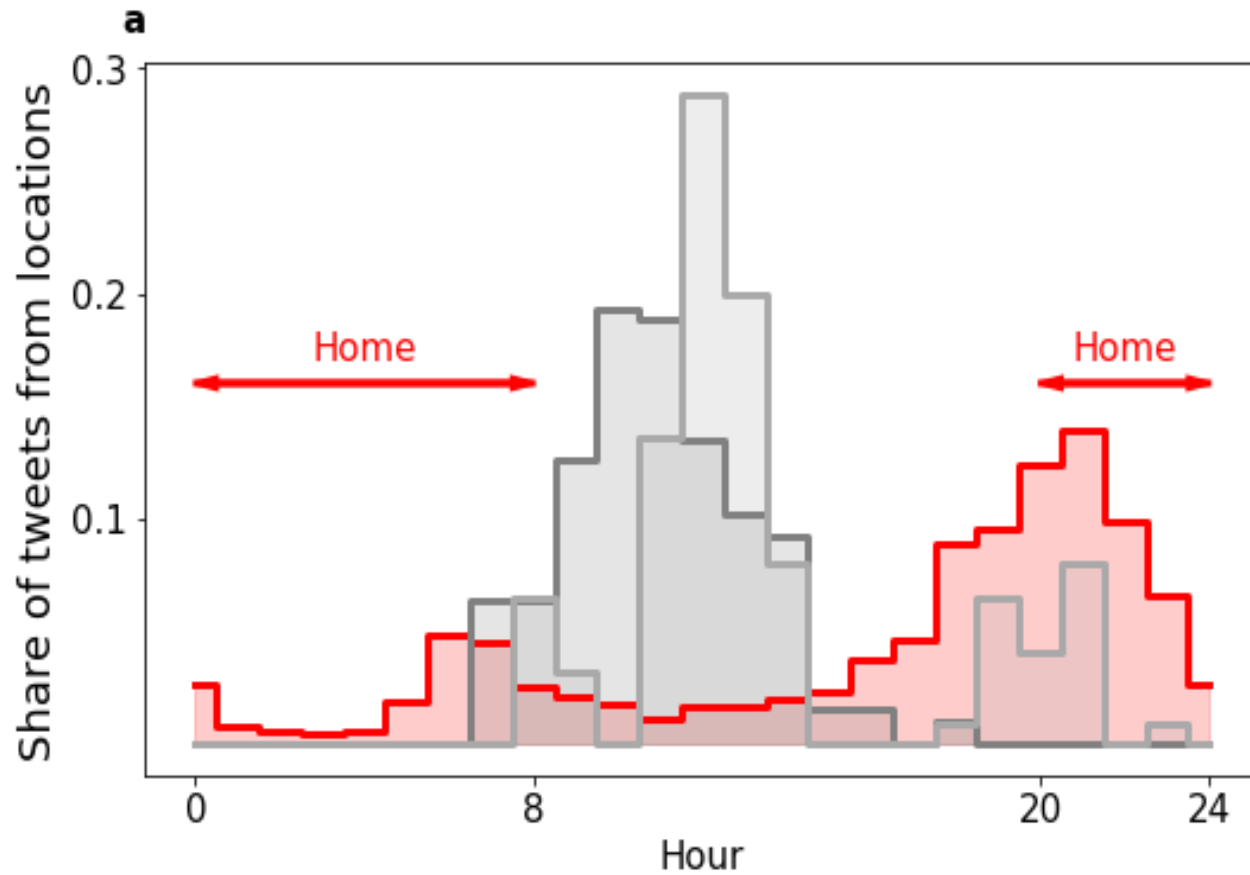
Geolocation
possibility

||| View Tweet activity

1% of all tweets and 200 tweets of a selected user available every day

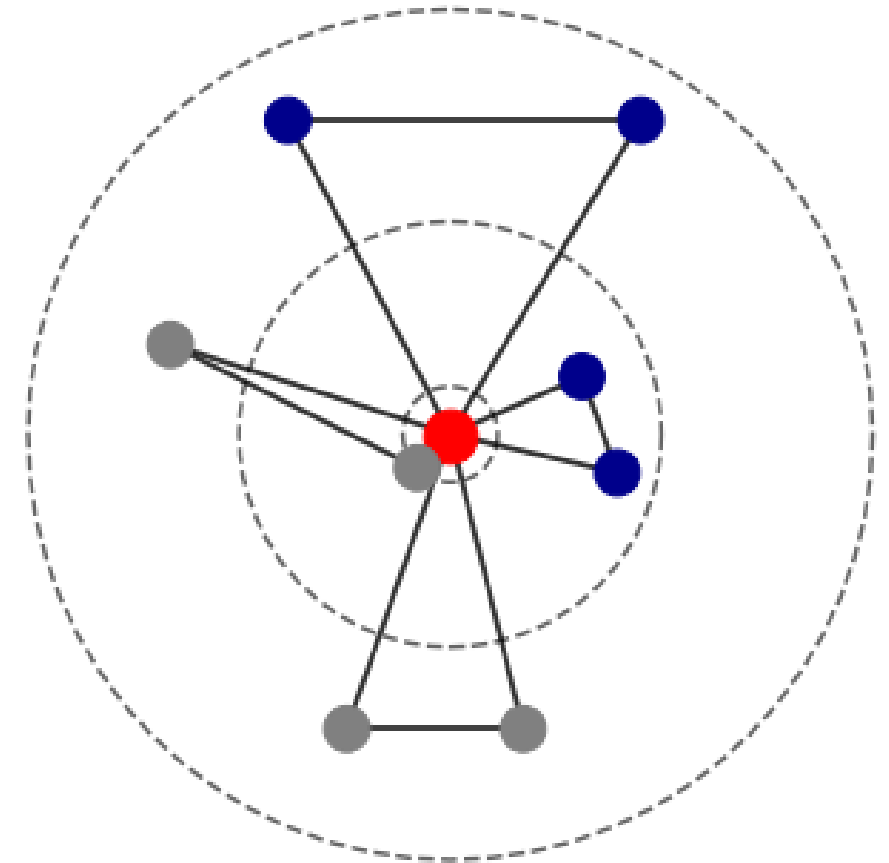
Focus on users with frequent geolocated tweets

Spatial social network of Twitter users



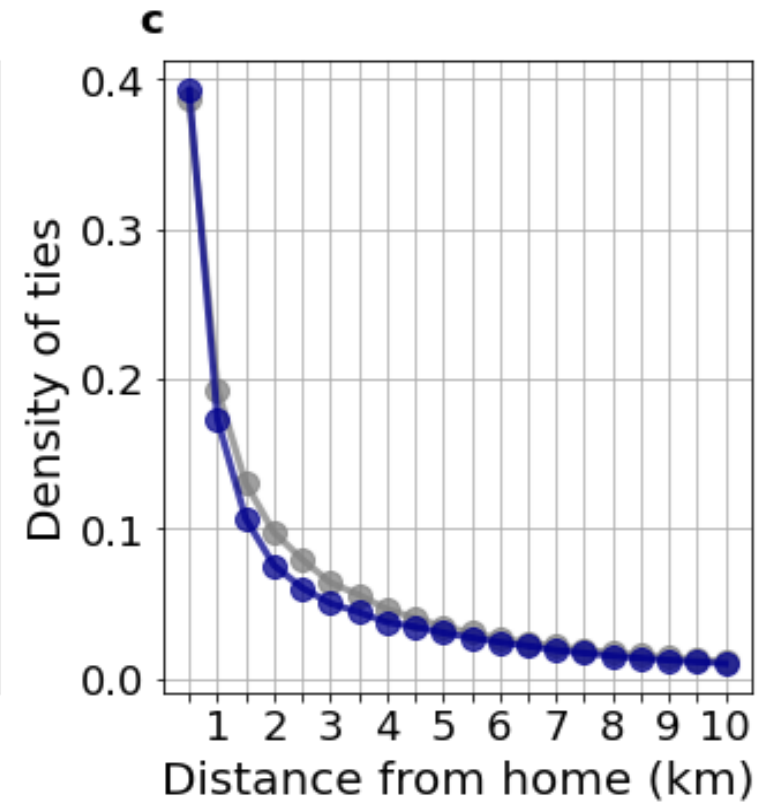
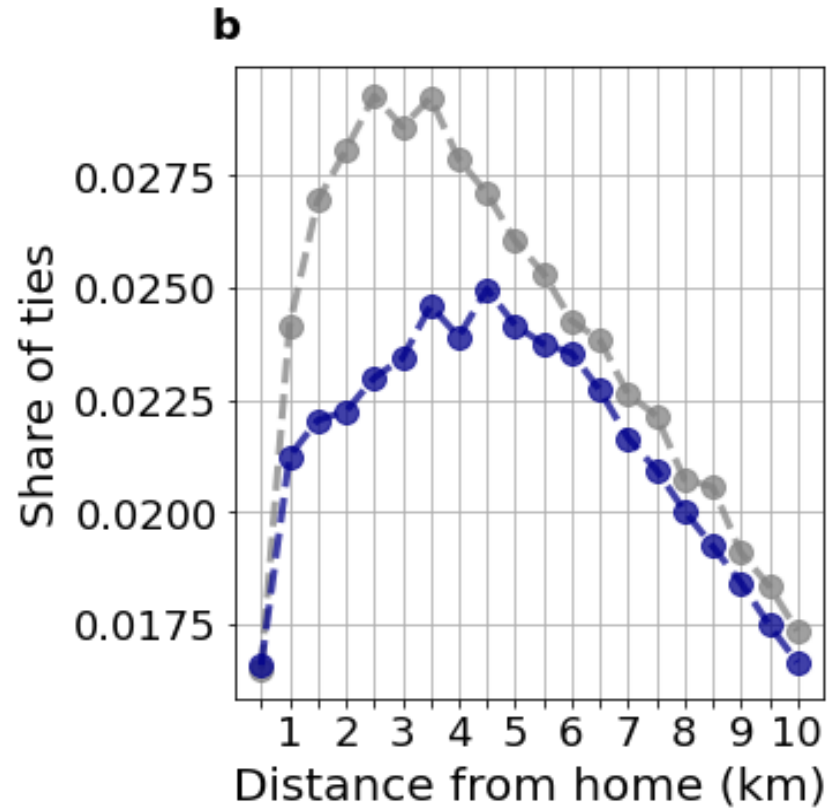
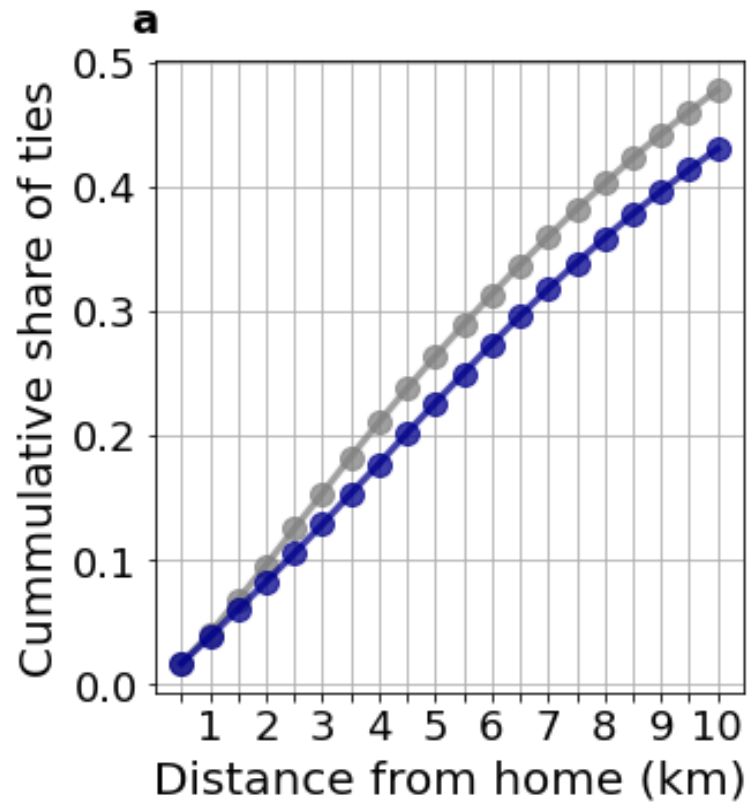
Key network measures

	Full graph	Income < median	Income > median
Degree in 10 km	8	4	4
Share of ties in 1 km	0.125	0.25	0
Share of ties in 5 km	0.375	0.25	0.5
Share of ties in 10 km	1	1	1
Clustering in 1 km	-	-	-
Clustering in 5 km	0.333	-	1
Clustering in 10 km	0.143	0.6	0.6
Support in 1 km	-	-	-
Support in 5 km	0.667	-	1
Support in 10 km	1	1	1

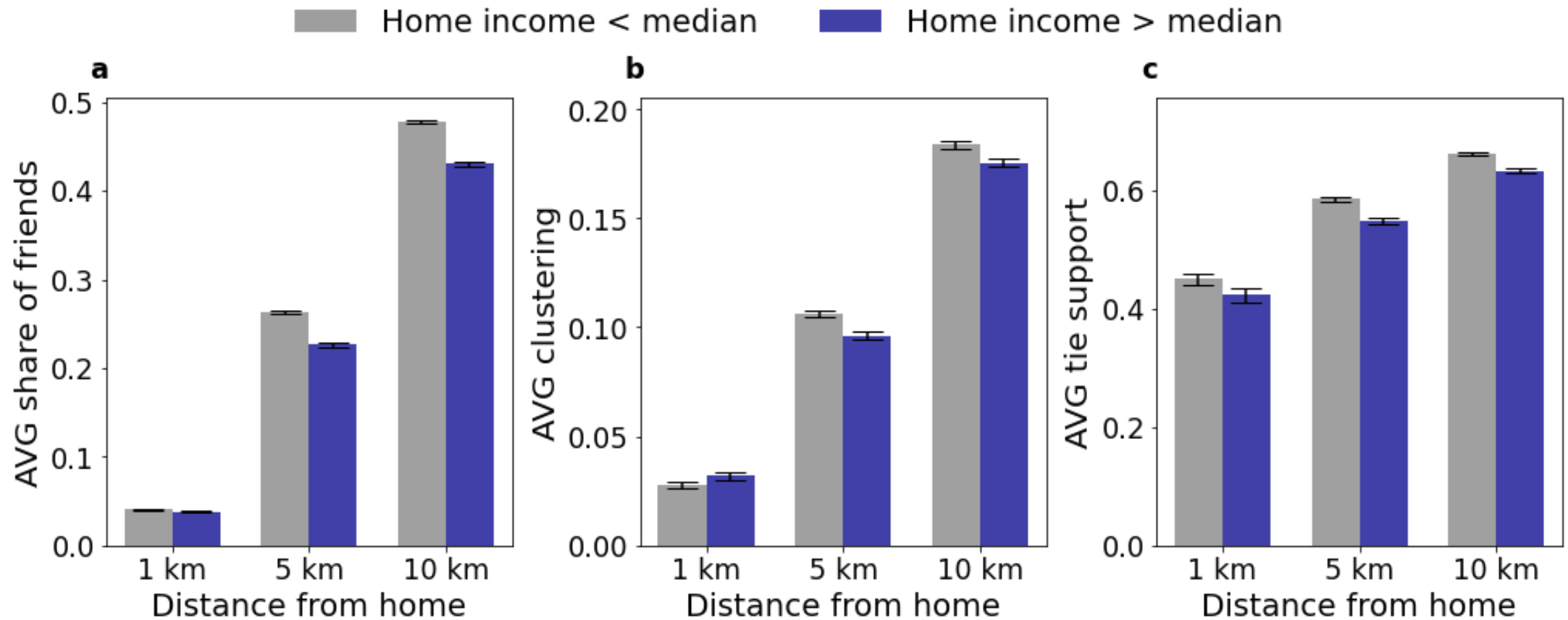


Concentration of social ties around home

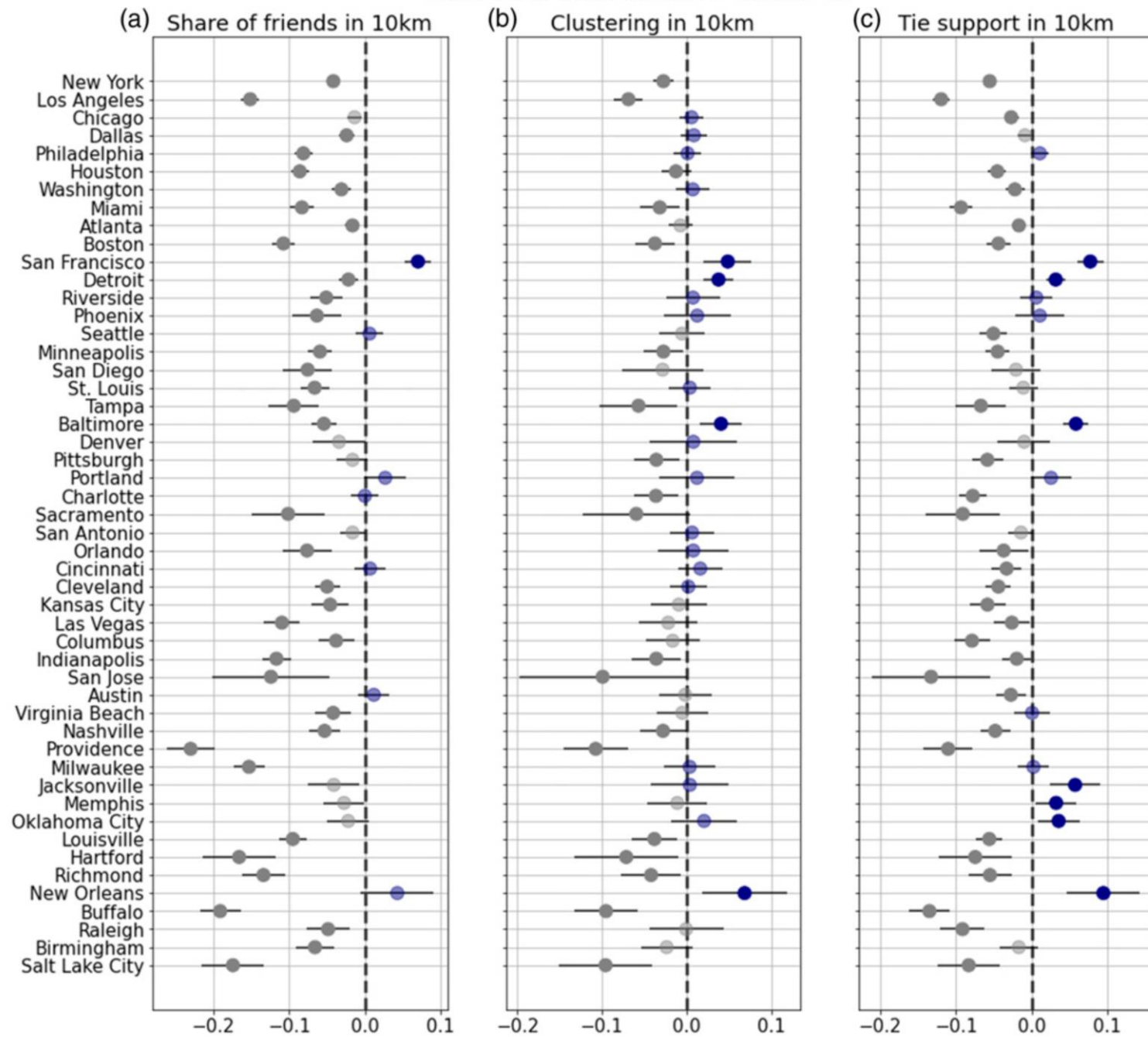
■ Home income < median ■ Home income > median



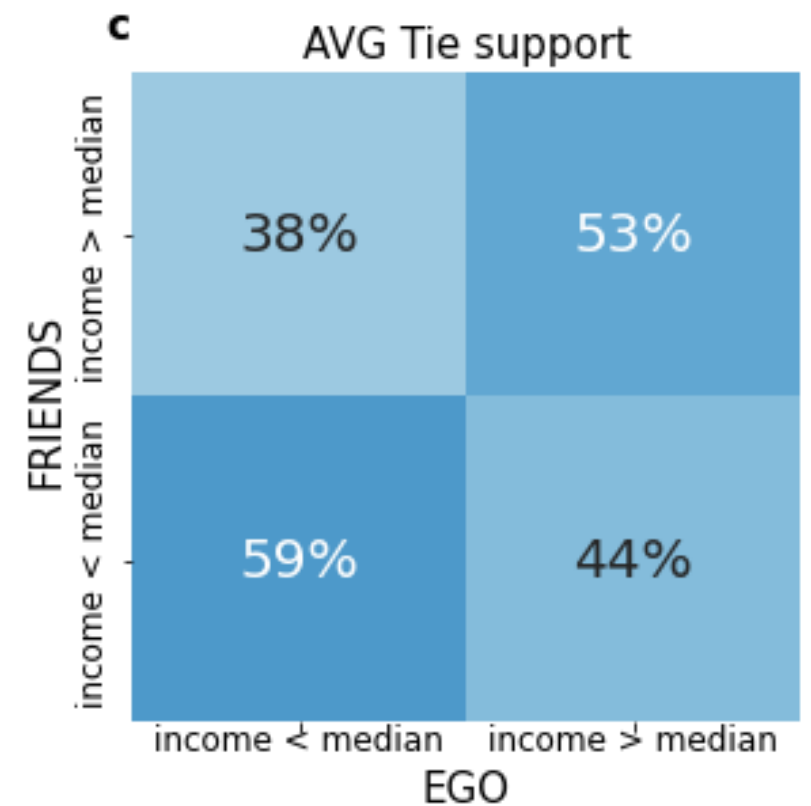
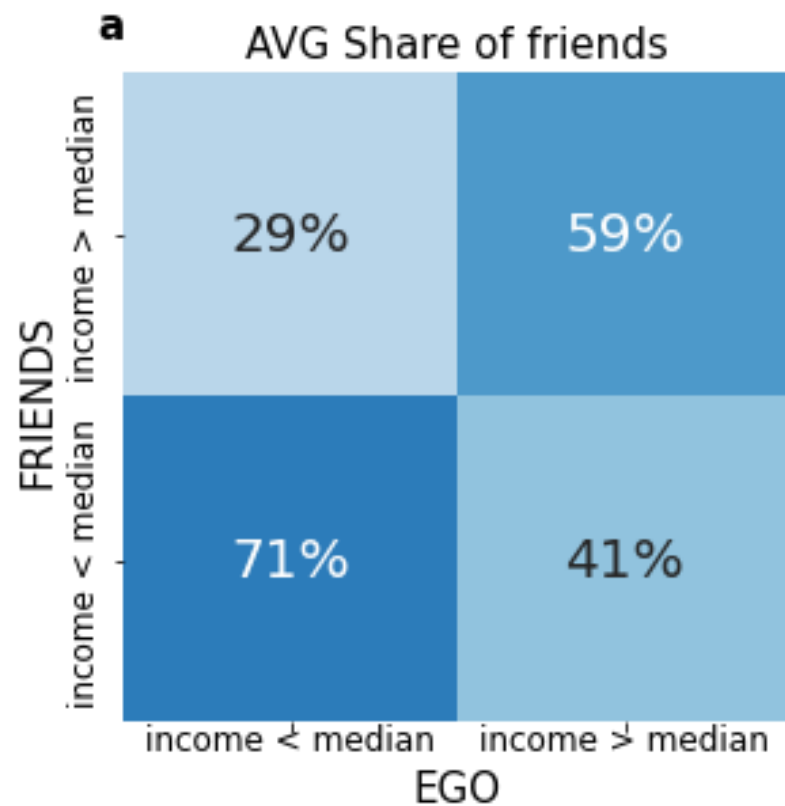
Concentration of social ties around home



Influence of home income > median on



Homophily of connections within 10 km



3.2) Universal patterns of long-distance commuting and social assortativity in cities

Home/work and spatial social network from Twitter messages

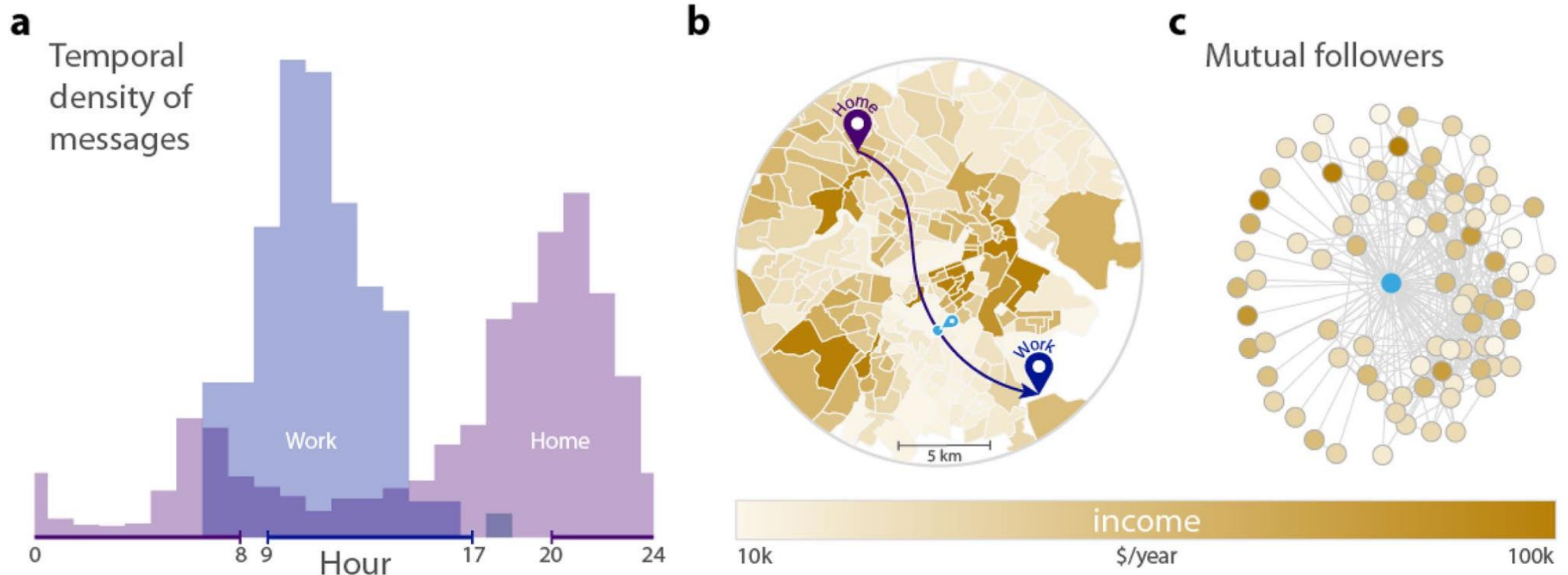
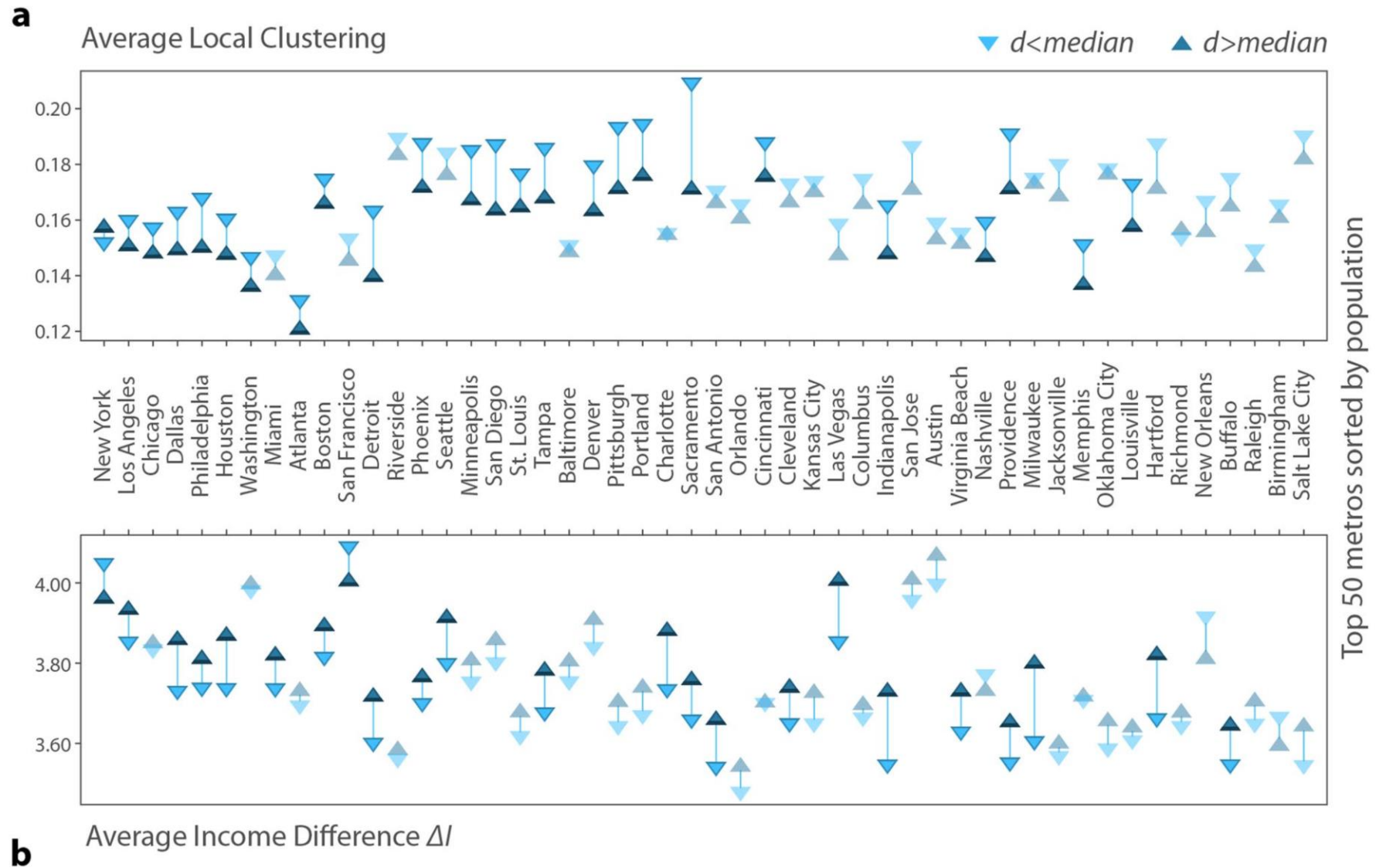
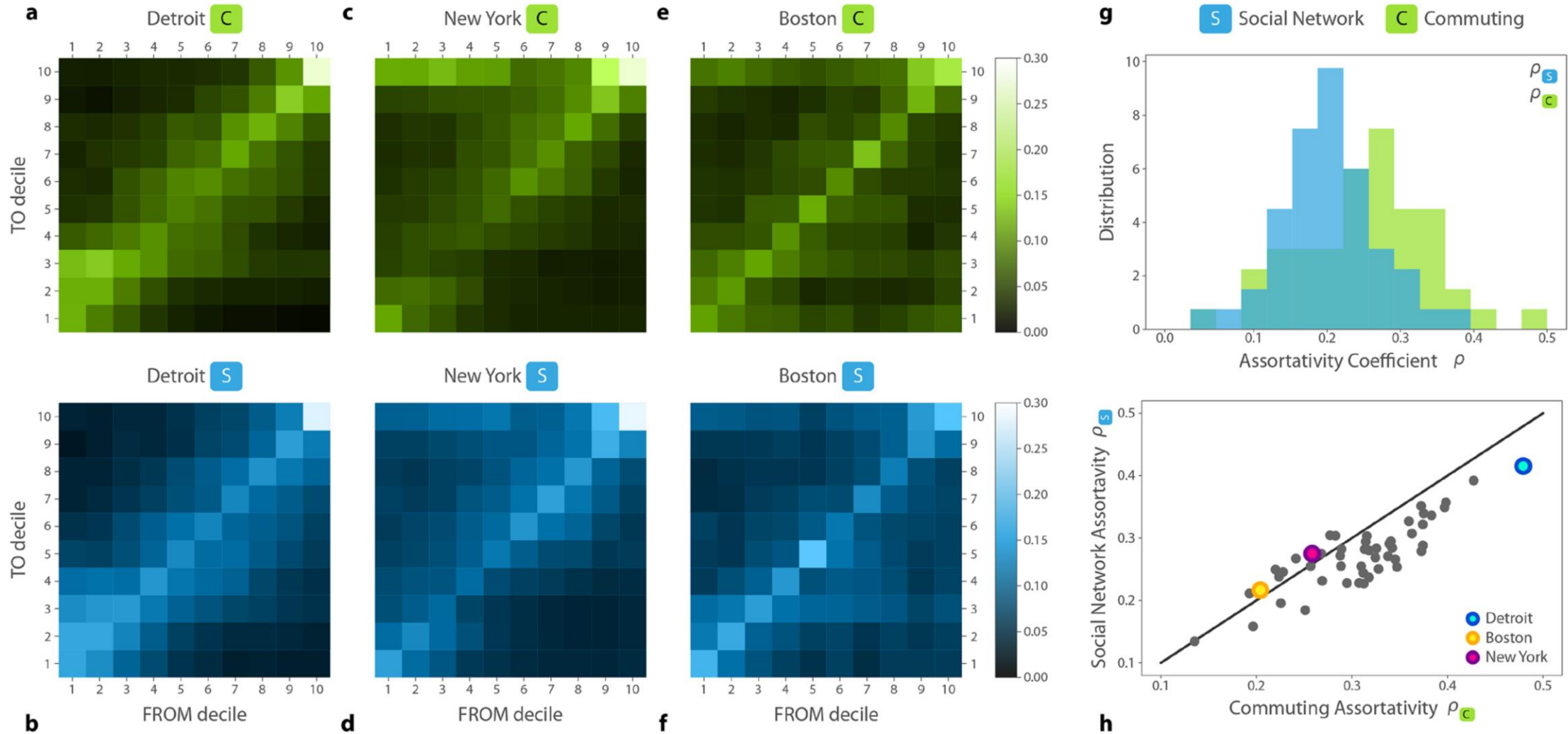


Figure 1. Combination of spatial, temporal and social network data of geolocated Twitter messages. (a) Home

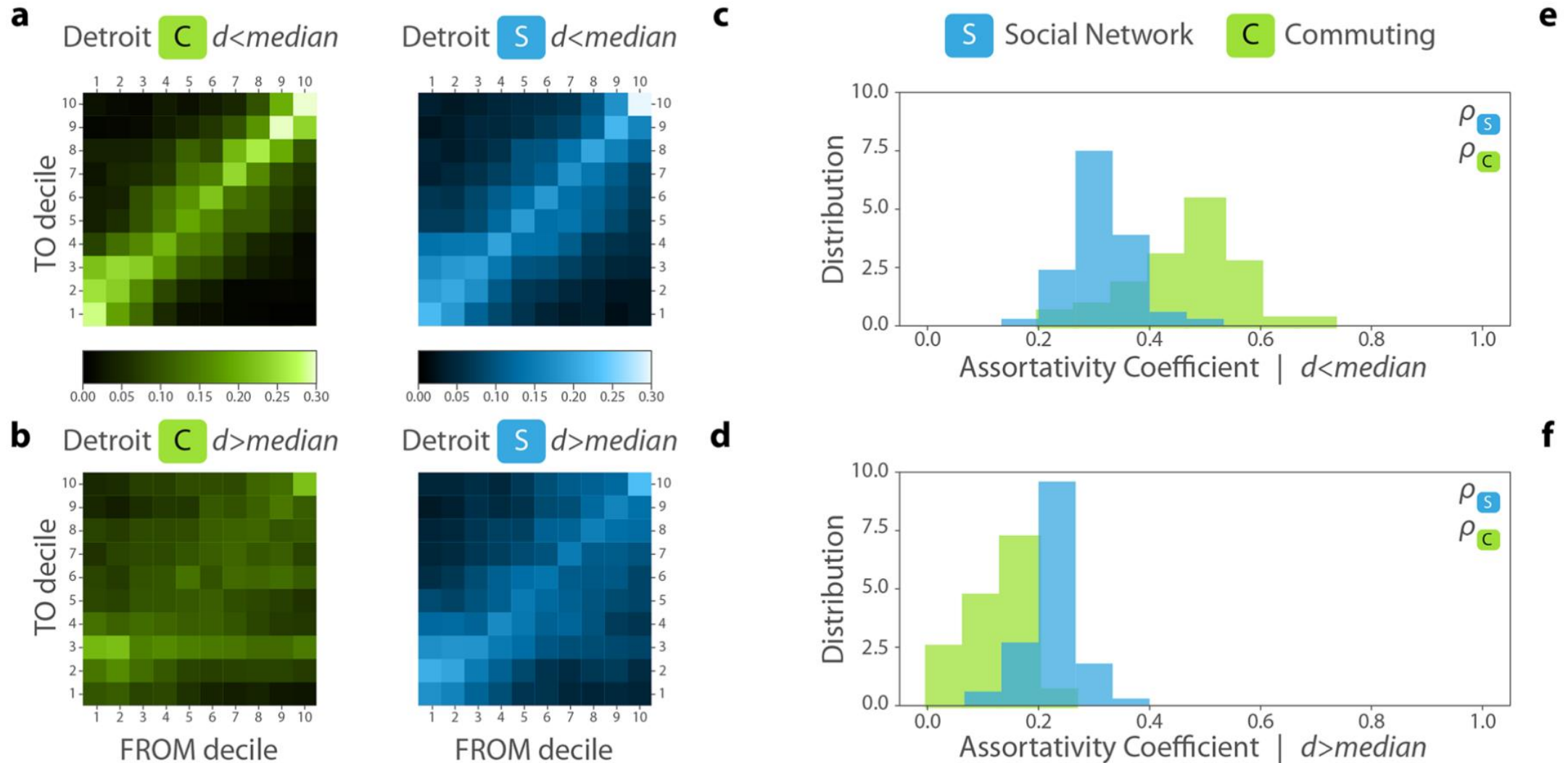
More movement, more open social network



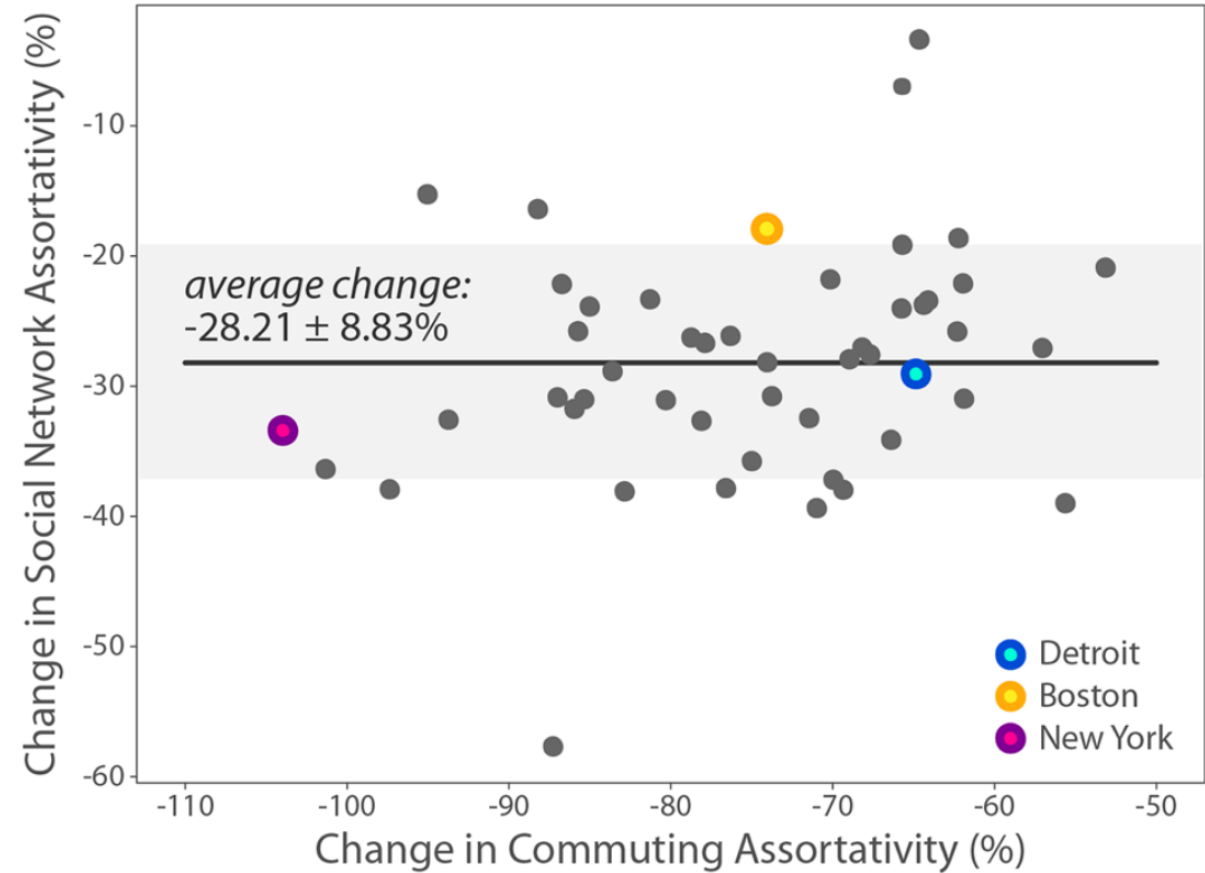
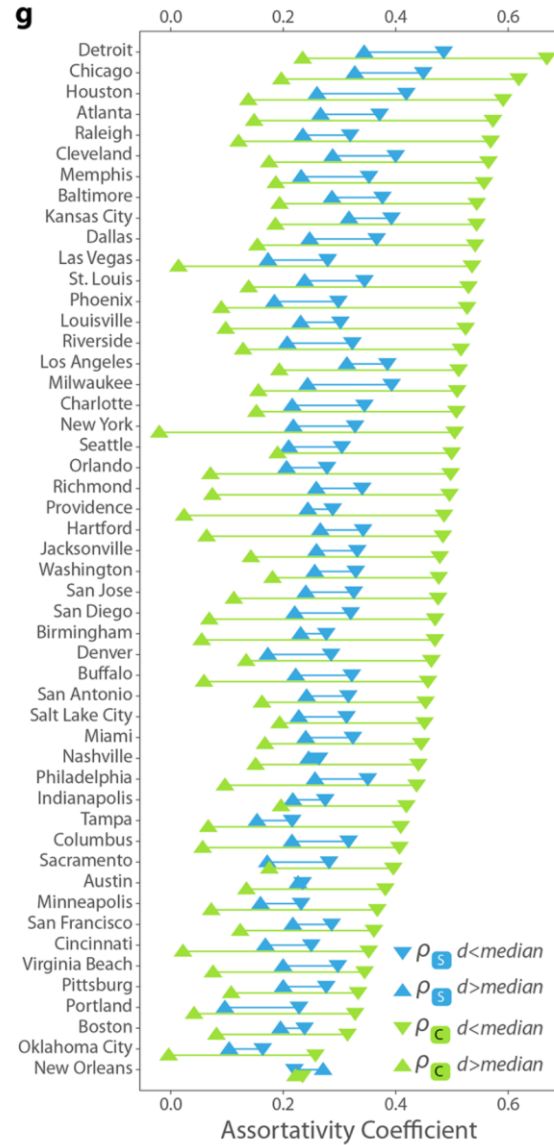
Assortativity in commuting and social relationships



Assortativity in commuting and social relationships



More commuting, lower assortativity of social ties

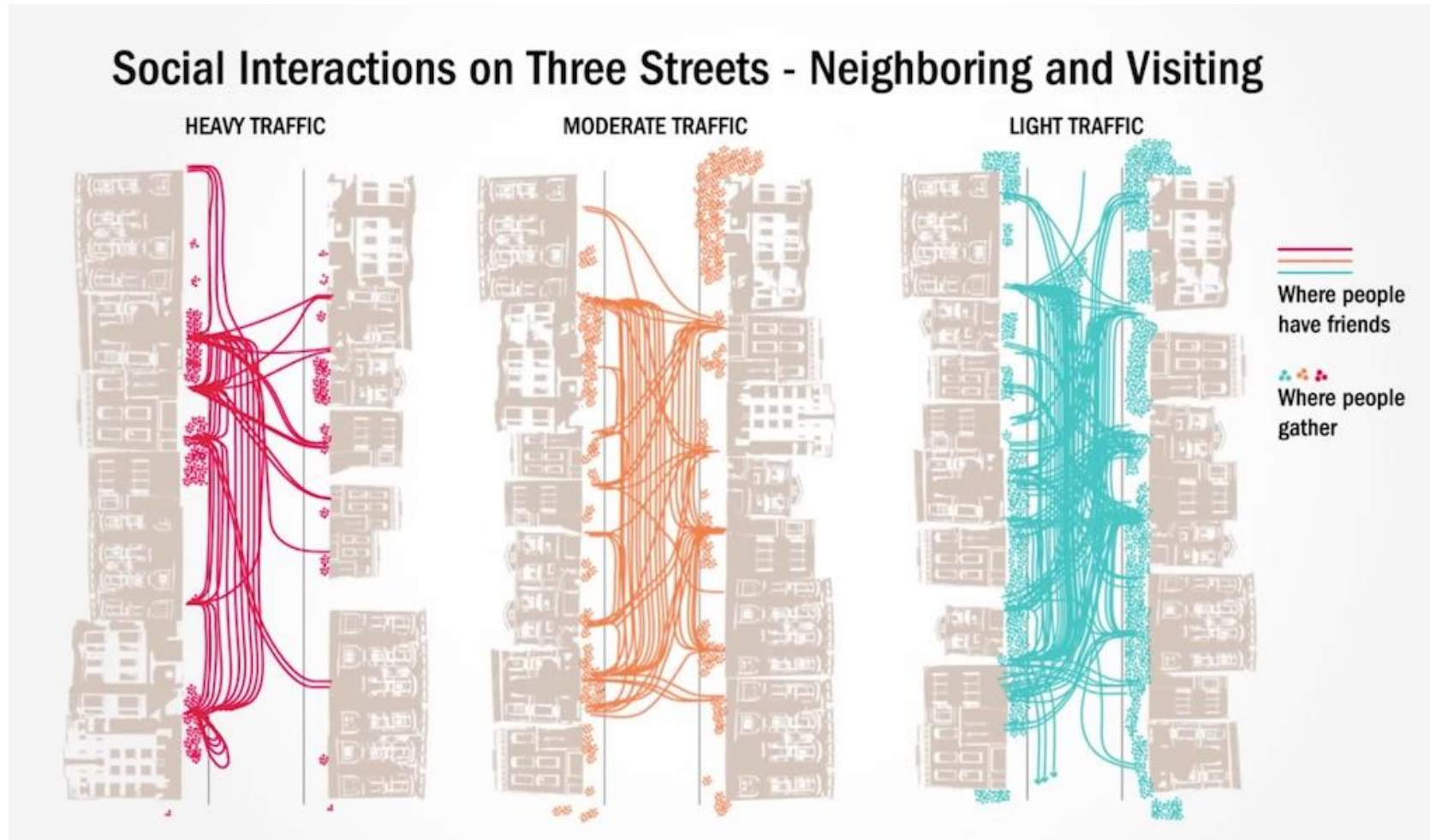


3.3) Urban highways are barriers to social ties

Connecting and dividing at the same time



Highways and social ties



Appleyard, D. (1981) Livable Streets

Highways and social ties

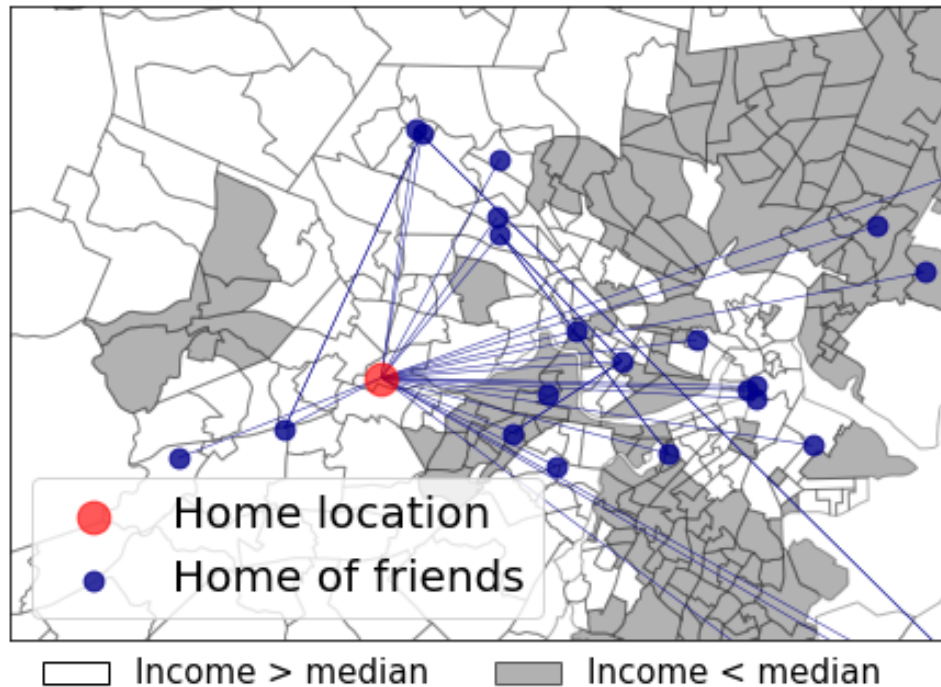
Large-scale
quantitative
evidence on how
highways interact
with social ties
within cities



Social ties

Twitter user activity (2012-2013)
1M users, 2.7M connections
in the top 50 US metropolitan areas

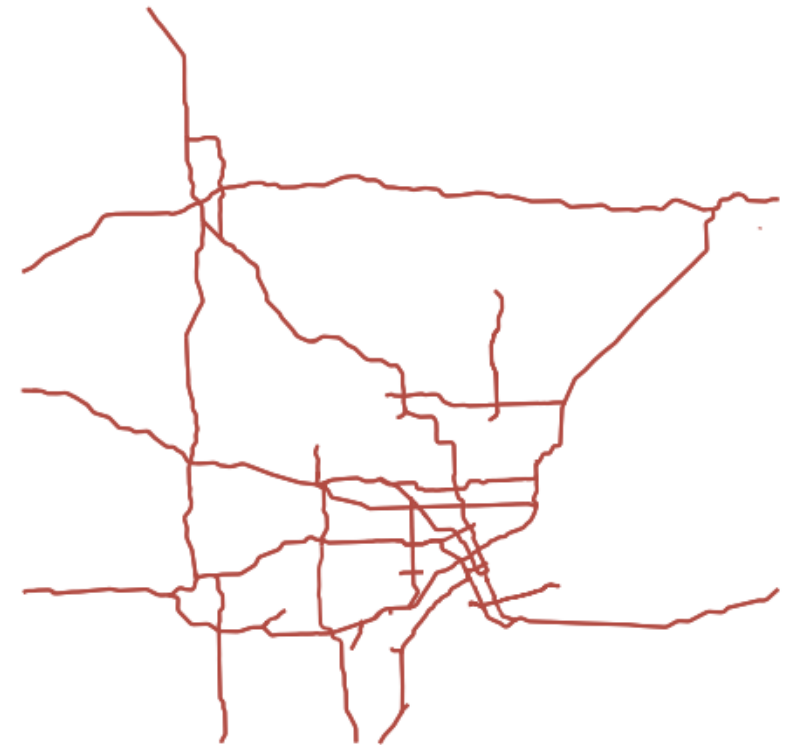
b



Kovács et al. (2022) **EPB**

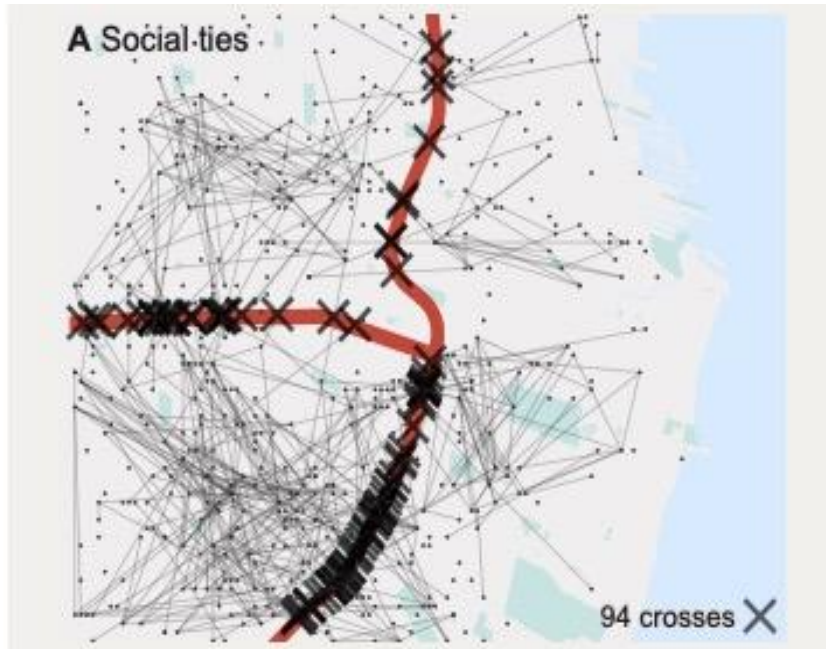
Highways

OpenStreetMap (OSM)
(*motorway + trunk links*)

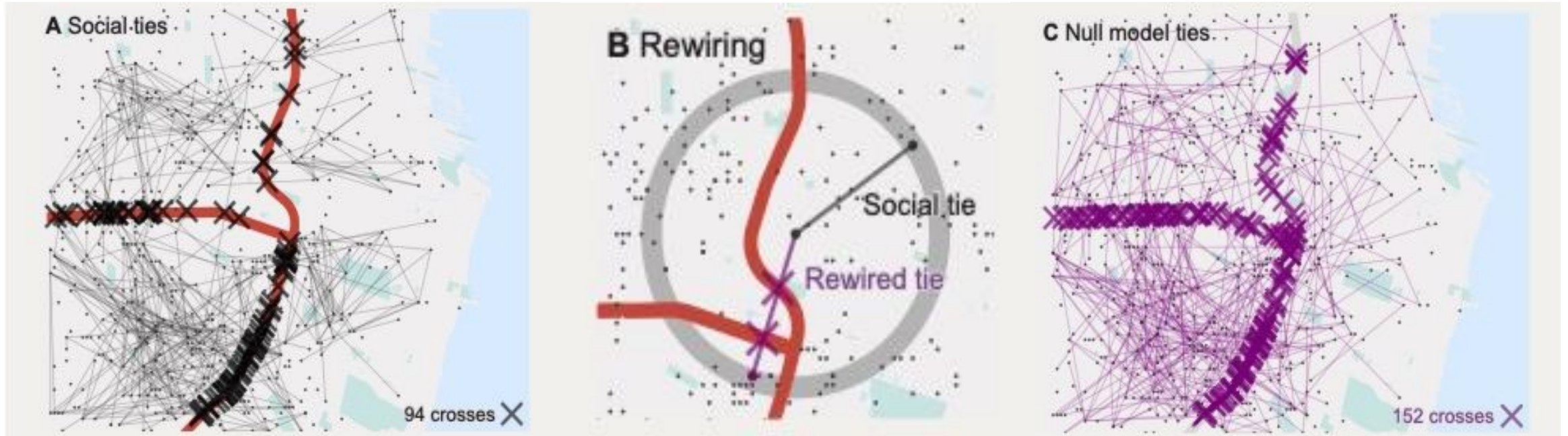


Detroit (-84.158, 42.028, -82.334, 43.327)

Barrier Score

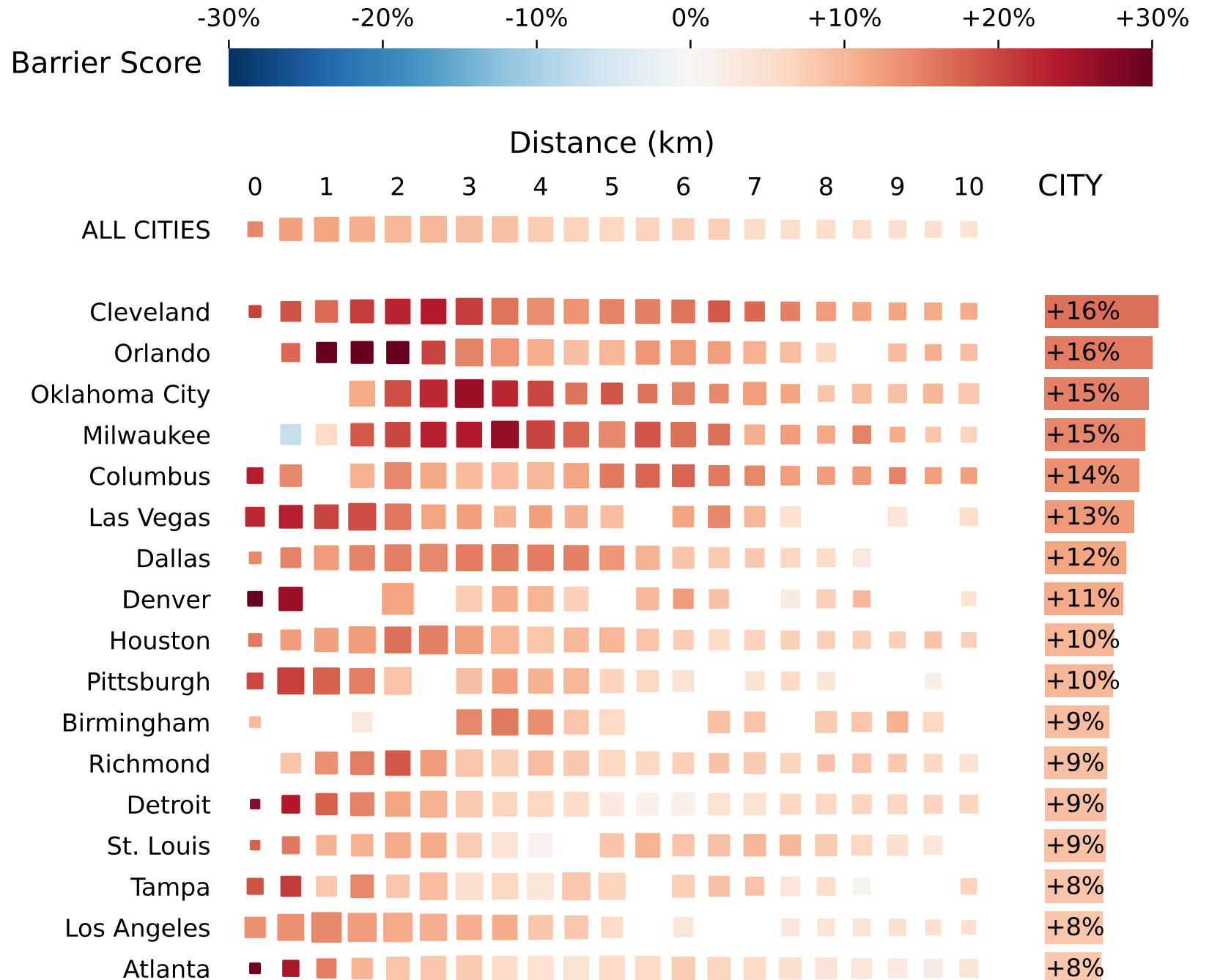
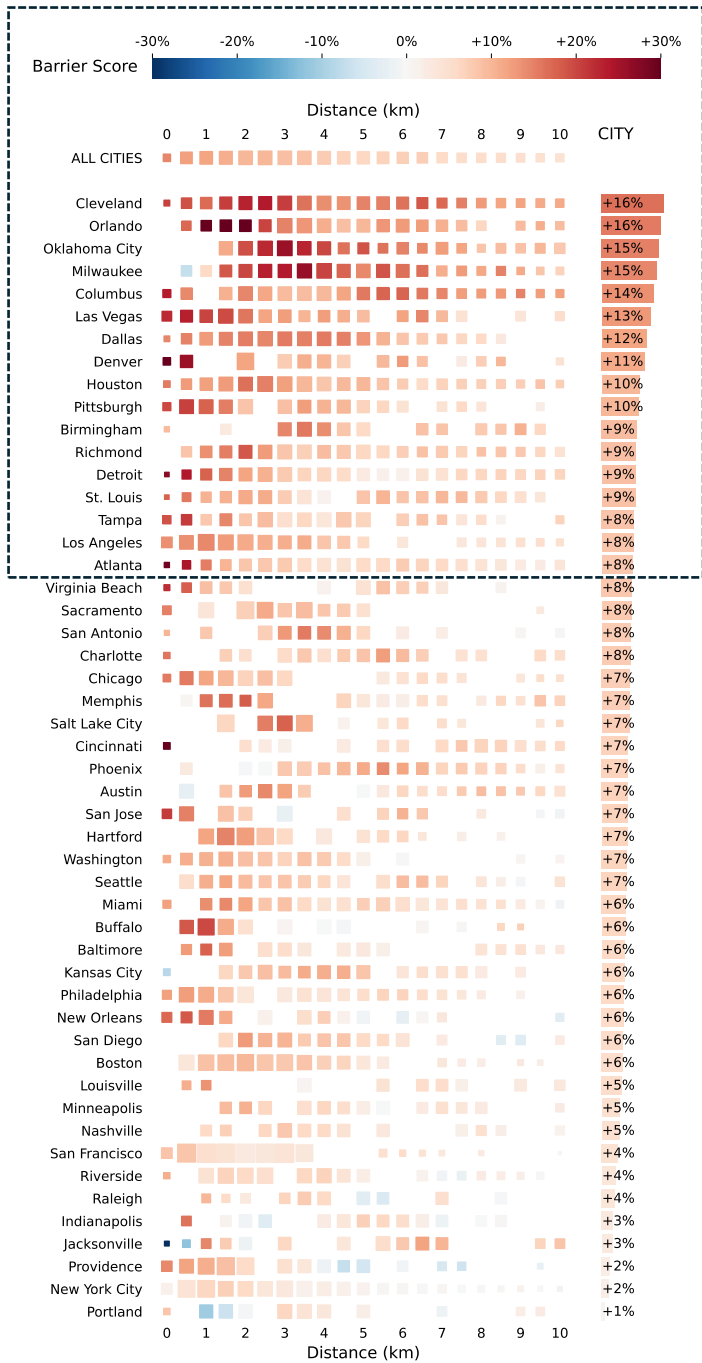


Barrier Score

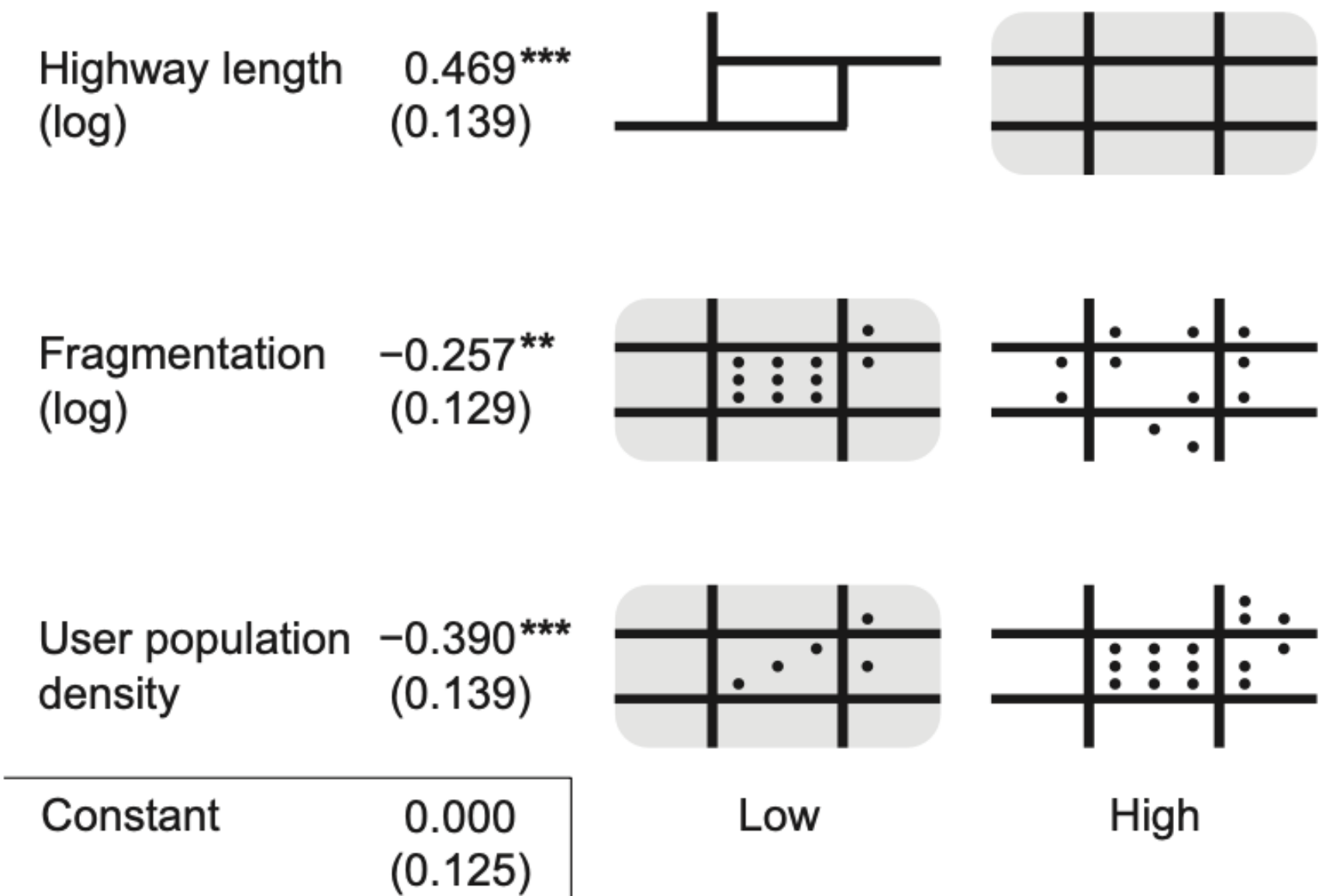


D Barrier Score

$$B_i = \frac{152-94}{94} = \mathbf{+62\%}$$

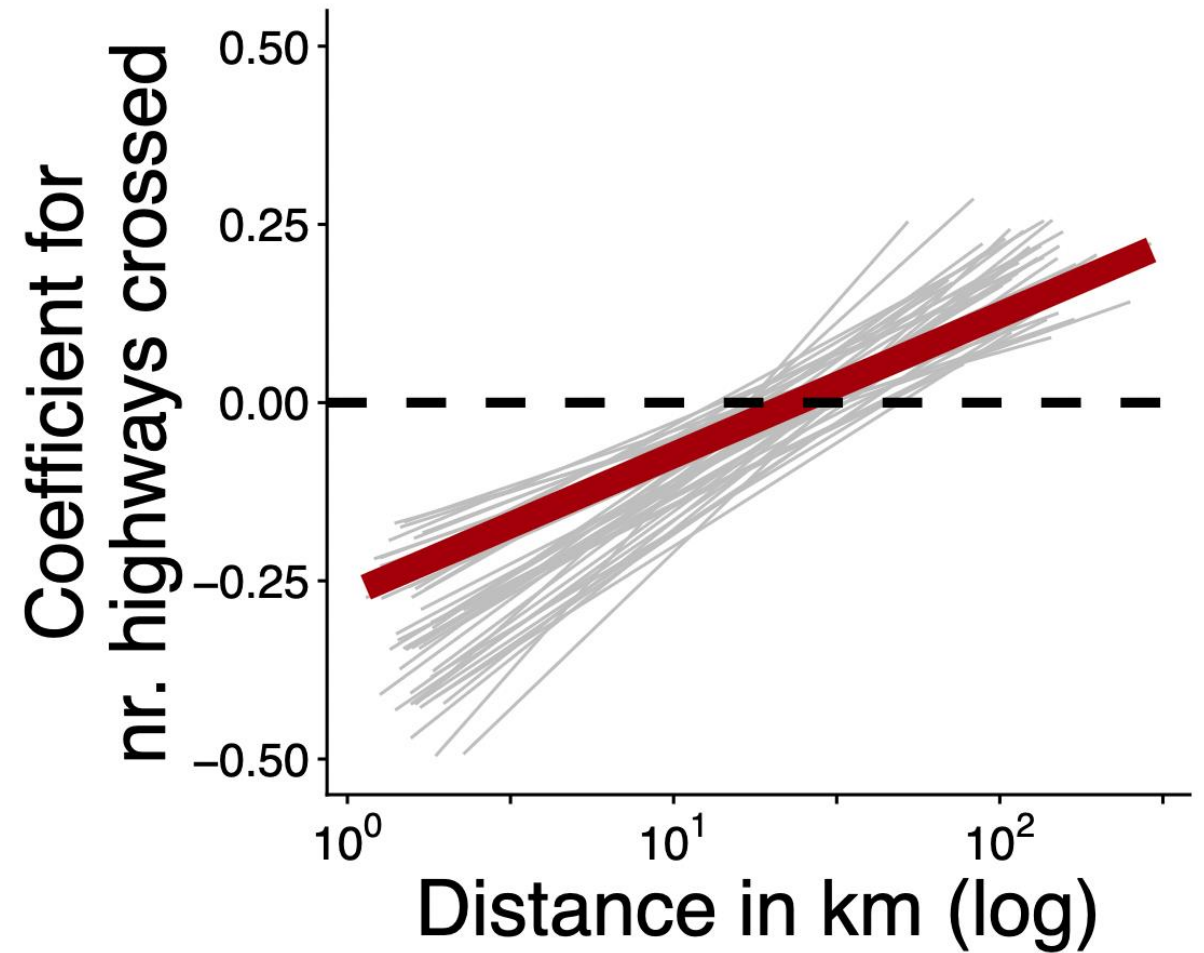


Barrier Score

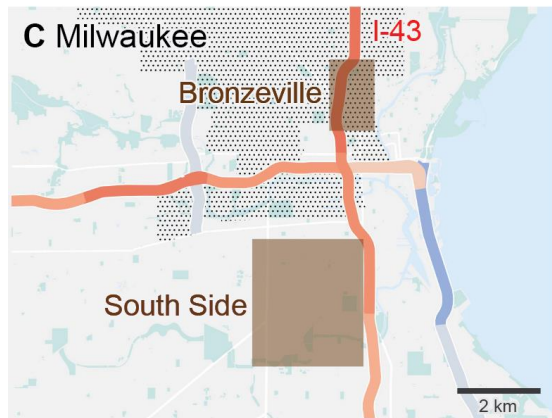
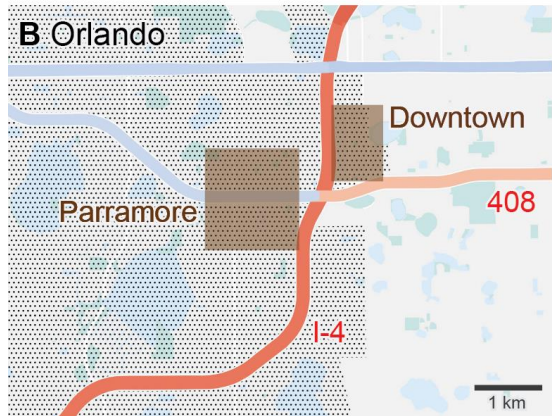
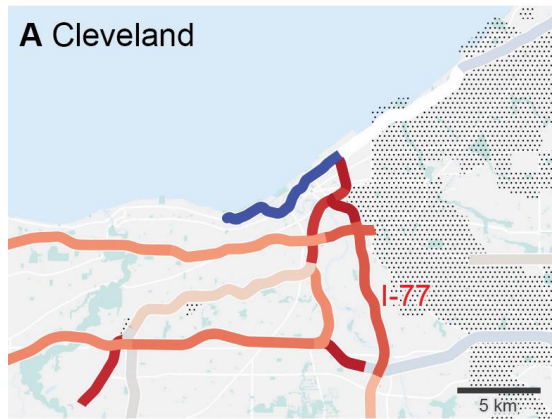


	Number of social ties (log)					
	(1)	(2)	(3)	(4)	(5)	(6)
Nr. of highways crossed (log)		−0.025*** (0.003)				−0.013*** (0.003)
Nr. of railways and waterways crossed (log)			−0.018*** (0.005)			−0.013*** (0.004)
Income abs. difference				−0.019*** (0.002)		−0.018*** (0.001)
Racial similarity					0.026*** (0.003)	0.023*** (0.003)
Distance (log)	−0.101*** (0.013)	−0.086*** (0.014)	−0.087*** (0.016)	−0.099*** (0.013)	−0.101*** (0.013)	−0.081*** (0.015)
User population (product log)	0.029*** (0.008)	0.027*** (0.007)	0.026*** (0.008)	0.026*** (0.008)	0.024*** (0.008)	0.019** (0.008)
Metro fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,669,688	2,669,688	2,669,688	2,669,688	2,669,688	2,669,688
R ²	0.042	0.043	0.043	0.045	0.047	0.051

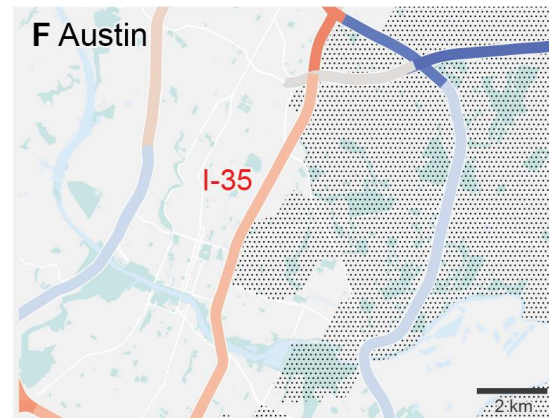
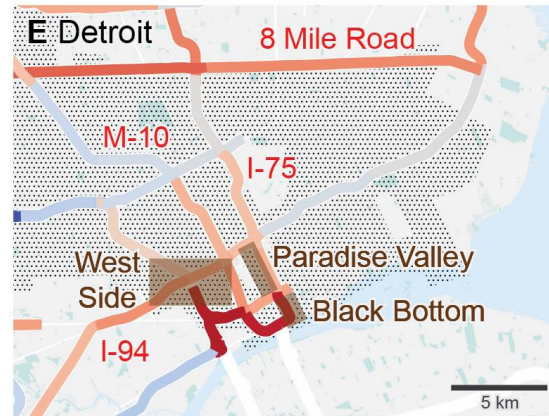
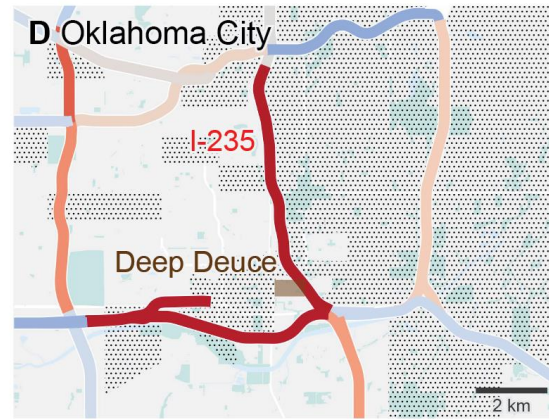
Estimator	OLS	OLS
Dependent variable	$\log(1 + t_{ij})$	$\log(1 + t_{ij})$
	(1)	(2)
Nr highways crossed (log)	-0.013*** (0.003)	-0.270*** (0.013)
Distance (log)	-0.081*** (0.015)	-0.238*** (0.013)
Distance X Nr highways crossed		0.197*** (0.008)
Nr. of railways and waterways crossed (log)	-0.013*** (0.003)	-0.013*** (0.004)
Income abs difference	-0.018*** (0.001)	-0.016*** (0.001)
Racial similarity	0.023*** (0.003)	0.021*** (0.003)
Population (product log)	0.019** (0.008)	0.018*** (0.007)
Metro fixed effect	Yes	Yes
Observations	2,669,688	2,669,688
R ²	0.051	0.064



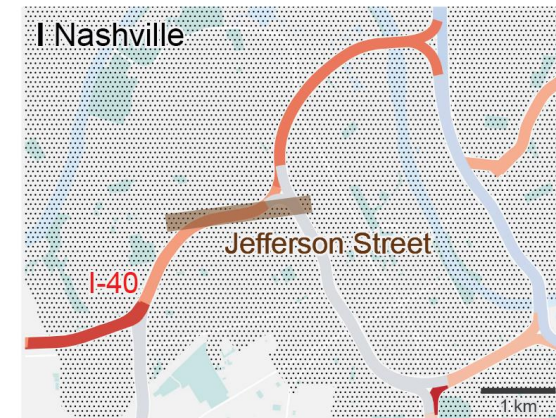
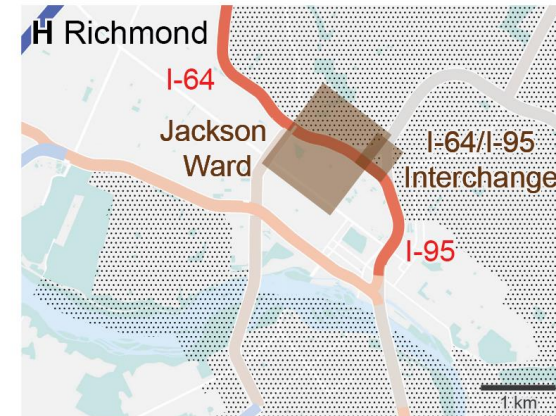
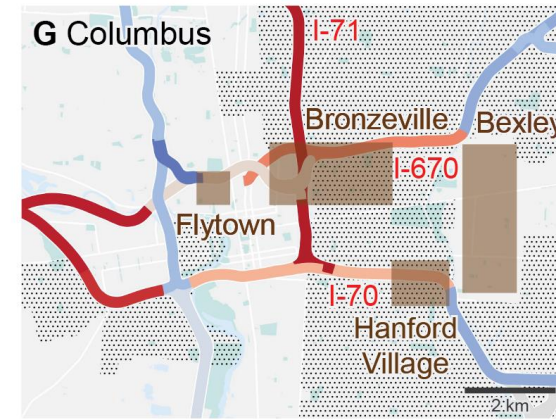
Top 3 Barrier Scores



Interracial Barriers

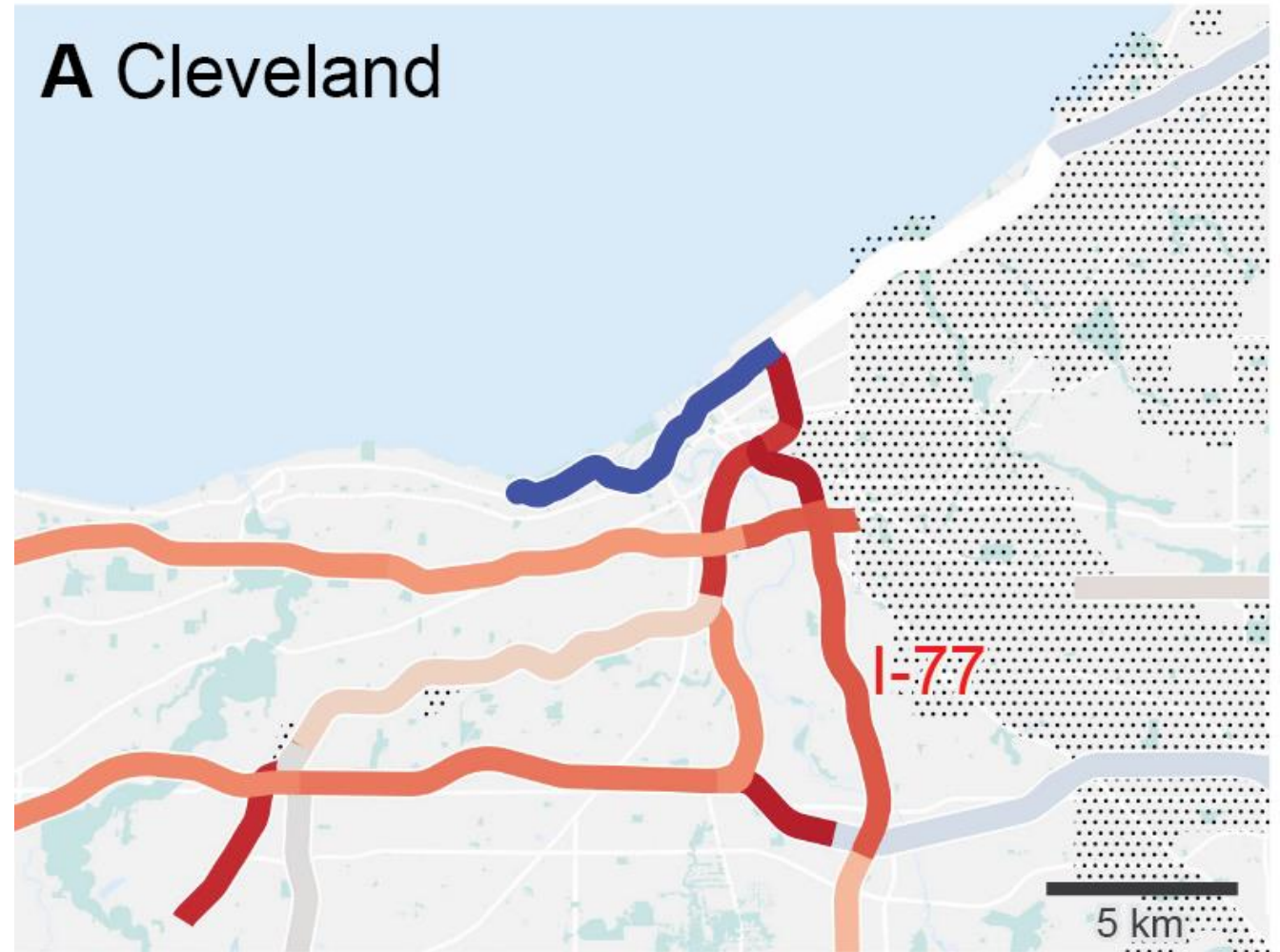


Intraracial Barriers



High Barrier Scores

I-77 separates the cluster of predominantly Black communities in the east from the rest of the metropolitan area



Area of interest

Black, upper quartile

Barrier Score:

-30%

-15%

0%

+15%

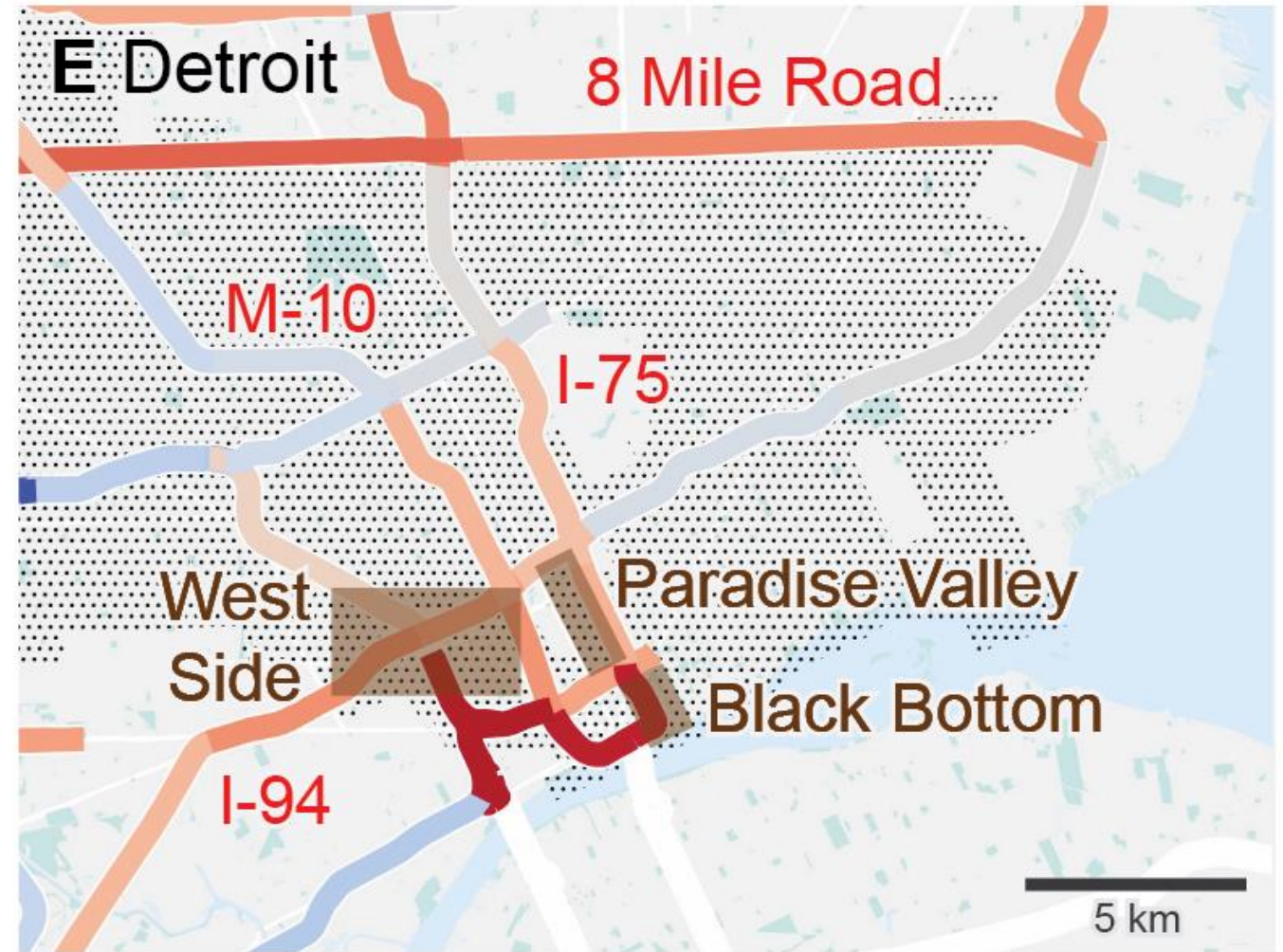
+30%

No data

Interracial Barriers

The Birwood Wall (Eight Mile Wall) manifesting the divide between the Black inner city and the white suburbs

Very high Barrier Scores in the inner city close to the historically Black neighborhoods



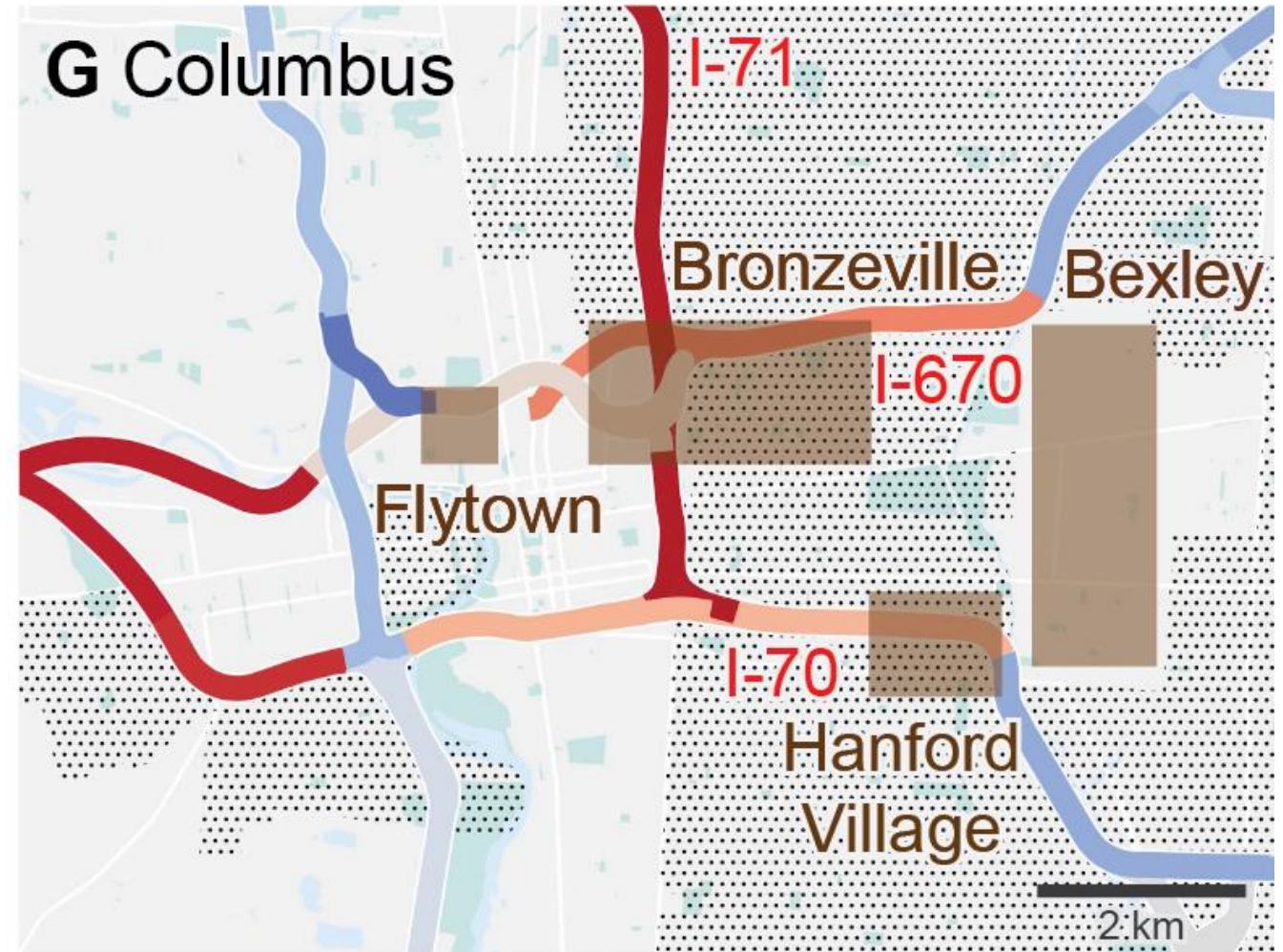
Area of interest

Black, upper quartile

Barrier Score: -30% -15% 0% +15% +30% No data

Intraracial Barriers

Highway construction as deliberate destruction of economically disadvantaged and predominantly Black neighborhoods



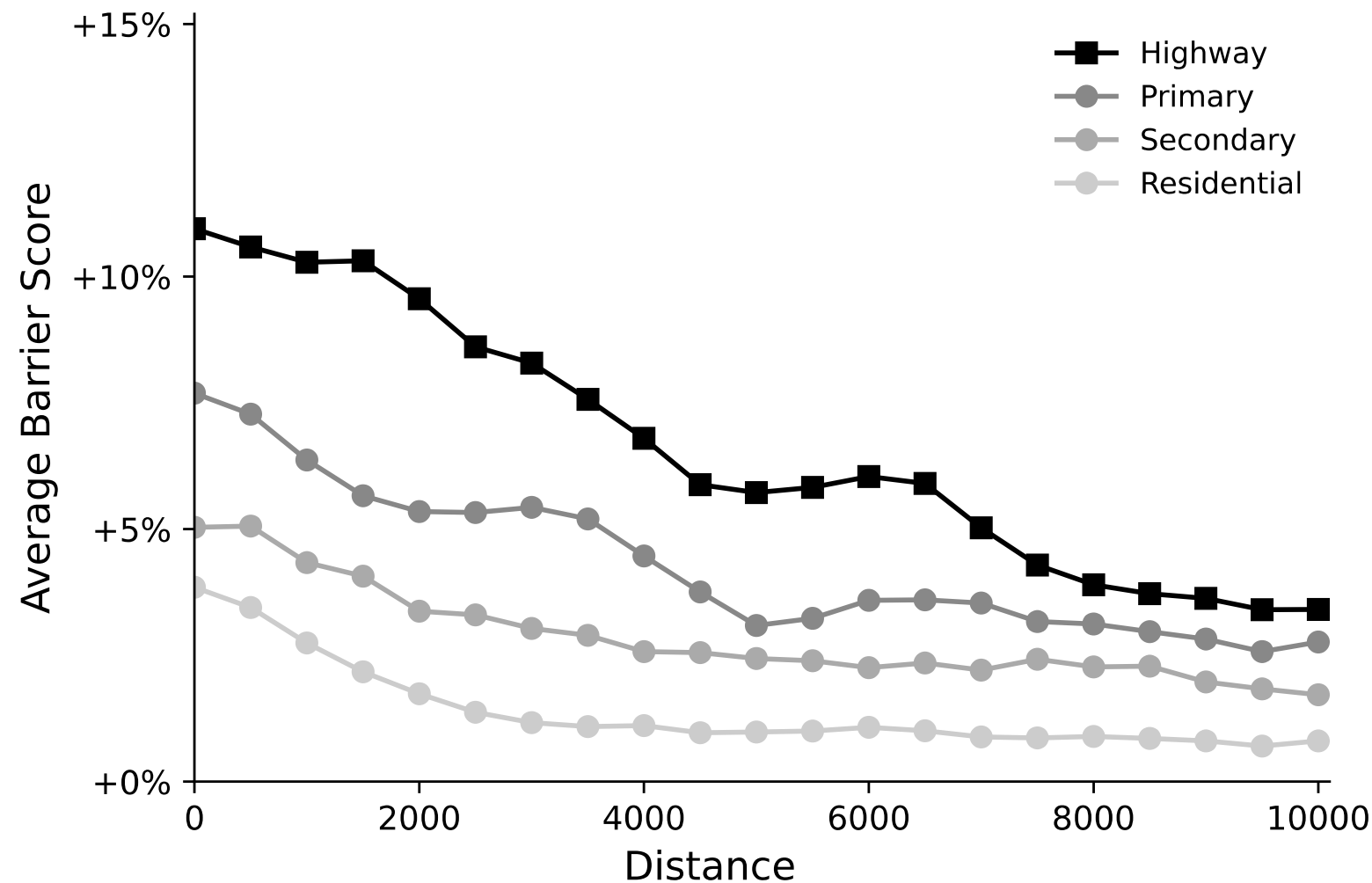
Area of interest

Black, upper quartile

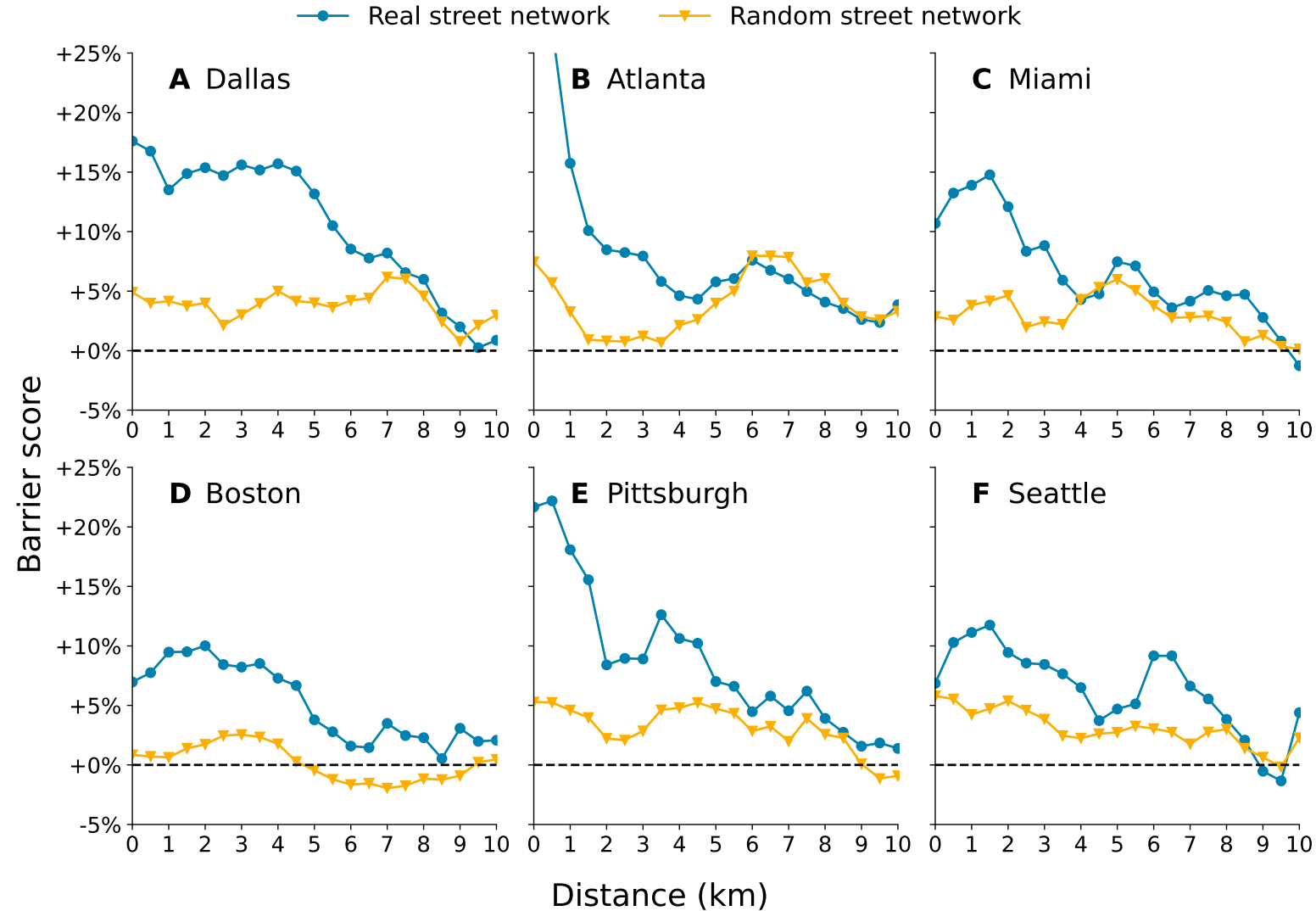
Barrier Score:



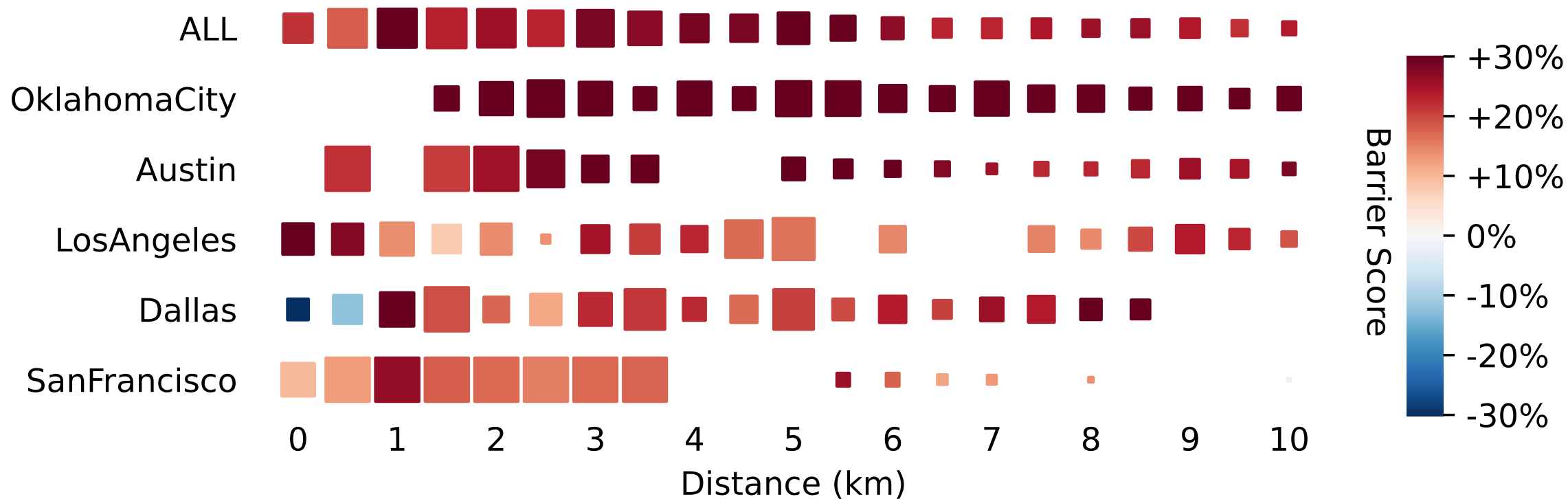
Robustness checks – other road categories



Robustness checks – random highway network



Robustness checks – Gowalla data



Discussion

Adding a highly granular perspective, we quantify the intuition that urban highways are barriers to social ties

Highways facilitate connections over longer distances, but at considerable harms with long lasting effects

Enrich the debate on contemporary highway policies and contribute to a more careful, evidence-based urban planning

Thank you!

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4. Homeworks and assignments